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Thermophysical properties of propan-2-ol (isopropyl alcohol):

Part 2. Thermodynamic properties at elevated pressures

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The Institution of Chemical Engineers

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THE PREPARATION OF THIS DATA ITEM

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Dr D.T. Jamieson	—	National Engineering Laboratory, East Kilbride, UK
Mr S.R. Knight	—	Stone & Webster Engineering Ltd, Milton Keynes, UK
Dr M. Schoenberg†	—	Hoechst AG, Frankfurt, Germany.

* Emeritus Member.

† Corresponding Member.

The technical work involved in the assessment of the available information and the construction and subsequent development of the Data Item was undertaken (under contract to ESDU) by

Dr I.J. Lawrenson	—	Division of Chemical Standards, National Physical Laboratory

with the assistance of Mr J. Walton.

The person with overall responsibility for the work in this subject area is Mr G.H. Walter, Head of Heat Transfer, Internal Flow and Physical Data Group.

THERMOPHYSICAL PROPERTIES OF PROPAN-2-OL (ISOPROPYL ALCOHOL): PART 2. THERMODYNAMIC PROPERTIES AT ELEVATED PRESSURES

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THERMOPHYSICAL PROPERTIES OF PROPAN-2-OL (ISOPROPYL ALCOHOL): PART 2. THERMODYNAMIC PROPERTIES AT ELEVATED PRESSURES

1. INTRODUCTION

This Item, which is one of a series concerned with the thermophysical properties of industrially important fluids, presents values of the thermodynamic properties of propan-2-ol for temperatures between 270 K and 580 K for pressures up to 50 MPa. Saturation data at fixed pressures are presented in Table 6.2 and data for the single phase in Table 6.3.

All quantities are expressed in SI units or their multiples (see Section 2) but, for users who require them in other units, conversion factors are provided in Appendix A. A list of the tabulated properties is given in Section 3, and Section 4 gives an explanation of the Tables in the Data Item.

A temperature-entropy chart is available from ESDU on request. Both the Tables and the Chart were generated by computer. Users who require saturation or ideal gas thermodynamic properties at fixed temperature, or data for surface tension, thermal conductivity (at saturation or in the dilute gas), or viscosity (at saturation or in the dilute gas) should refer to the companion Item, Data Item 89017.

2. NOTATION AND UNITS

Symbolic equations are written in terms of thermodynamic temperatures, T. For most purposes the thermodynamic scale is interchangeable with the International Practical Temperature Scale of 1968 (Reference 1) and all numerical values are given as International Practical Kelvin Temperatures, T_{68}.

B	denotes $v - rT/p$	m^3/kg
c	specific heat capacity	kJ/kg K
h	specific enthalpy	kJ/kg
M	molar mass	kg/mol
p	pressure	kPa
R	molar (universal) gas constant	J/mol K
r	specific gas constant for isopropyl alcohol	J/kg K
s	specific entropy	kJ/kg K
T	kelvin temperature	K
T_{68}	International Practical Kelvin Temperature	K
V	molar volume	cm^3/mol
v	specific volume	m^3/kg
Z	compression factor, $pv/(rT)$	–
ρ	density	kg/m^3
ω	acentric factor	–

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State descriptors

(g)	value for gas or saturated vapour
(l)	value for single phase or saturated liquid
Δ	denotes property difference between two states

Superscripts

id	denotes ideal gas

Subscripts

b	value at normal boiling point
c	value at critical point
m	value at normal melting point
p	value at constant pressure
sat	value at saturation (defined in Appendix B, Section B1)
v	value at constant volume
vap	denotes evaporation.

3. THERMODYNAMIC PROPERTIES

The thermodynamic properties have been calculated from a single correlating function, a Helmholtz Equation of State, in a manner similar to that used to obtain tables of thermodynamic properties for steam (Reference 2) and toluene (Reference 3). The function chosen represents the Helmholtz energy, and other thermodynamic properties are obtained from this function by differentiation and by the application of thermodynamic relationships. The various property values so obtained are self-consistent and are calculated on the basis that the enthalpy and entropy of the crystalline solid are zero at 0 K. The function describing the Helmholtz energy contains a total of 53 constants fitted by a least-squares method to 761 critically selected experimental data points (either temperature-volume-pressure or temperature-volume-enthalpy). A more detailed description of the fitting procedure and of the selection of data will be found in Reference 3.

The following properties of the saturated fluid are given in Table 6.2 at selected values of pressure:

(i) temperature of saturation,

(ii) density of the saturated liquid,

(iii) specific volume of the saturated vapour,

(iv) specific enthalpy of the saturated liquid, specific enthalpy of the saturated vapour and specific enthalpy of vaporisation,

(v) specific entropy of the saturated liquid and specific entropy of the saturated vapour,

(vi) specific heat capacity at saturation of the saturated liquid.

The following properties of the liquid and vapour phase are given in Table 6.3 at selected pressures and at temperatures up to 580 K:

(i) specific volume,

(ii) compression factor,

(iii) specific entropy and enthalpy,

(iv) specific heat capacities at constant pressure and at constant volume,

together with these properties for the saturated liquid and saturated vapour where applicable.

Some of the data in the Tables have been represented graphically as a temperature-entropy chart including the region of saturation together with lines of constant pressure, constant volume and constant enthalpy. This chart is available from ESDU on request.

4. EXPLANATION OF THE TABLES

4.1 Derivation and Accuracy

The experimental data from the derivations listed in column 2 of Table 4.1 were used as input for the development of the Helmholtz Equation of State. The values quoted in the fourth column represent the maximum or the r.m.s. difference between the calculated values given in the Tables and the original, measured data.

TABLE 4.1
Data used in the development of the Helmholtz Equation of State

Property	Derivation	Temperature or pressure range	Deviation from Tables 6.2 and 6.3	Remarks
T_c	14			
p_c	14			
v_c	14			
p_{sat}	10	270 – 508 K		
$\rho_{sat}(l)$	12	270 – 430 K	0.03% (max.)	
$h_{sat}(l)$	6	270 – 330 K	0.04 kJ/kg (max.)	
	4	270 – 470 K	0.05 kJ/kg (max.)	
$\Delta_{vap}h$	5	324 – 355 K	0.04 kJ/kg (max.)	
	8		0.04 kJ/kg (max.)	
c_p^{id}	11	273 – 1000 K		The ideal gas part of the Helmholtz Equation of State was obtained by integration of the equation for c_p^{id}.
pvT properties in single phase region	13	385 – 573 K 2.7 – 13 MPa	0.51% (r.m.s.)	
	9	373 – 573 K 12 – 55 MPa	0.62% (r.m.s.)	
	7	373 – 473 K 0.2 – 2.2 MPa	0.37% (r.m.s.)	
		340 – 470 K 1 kPa	0.008% (r.m.s.)	Calculated from *B* defined in Section 2.
		340 – 470 K 10 kPa	0.07% (r.m.s.)	

4.2 Use and Interpolation

Tabulated values for the saturation properties at fixed pressures are given in Table 6.2. In Table 6.3 for isobars below the critical pressure, the first two entries are properties for the saturated liquid and vapour respectively. The remaining entries are properties for the single phase. Where the single phase temperatures, T_{68}, are below the saturation values, the properties refer to the liquid phase; the entries at temperatures above saturation refer to the vapour phase. Along isobars above the critical pressure, no liquid-vapour phase boundary appears at any temperature and the properties vary continuously over the whole temperature range.

In general the values given in Tables 6.2 and 6.3 are quoted to at least one significant figure more than is strictly warrented by the data accuracy to reduce errors in interpolation. Linear interpolation between adjacent tabulated values will usually be sufficient and, in Table 6.2, will not introduce errors greater than those indicated in Table 4.2.

TABLE 4.2
Maximum errors due to linear interpolation of the data in Table 6.2

Saturation property	Temperature range K	Maximum error due to linear interpolation
p_{sat}	$270 \le T \le 340$	3%
	$340 < T \le 390$	1.0%
	$390 < T \le 450$	0.5%
	$450 < T \le 500$	0.2%
$\rho_{sat}(l)$	$270 \le T \le 400$	0.01%
	$400 < T \le 440$	0.03%
	$440 < T \le 505$	0.6%
$v_{sat}(g)$	$270 \le T \le 300$	3%
	$300 < T \le 360$	2%
	$360 < T \le 420$	1%
	$420 < T \le 500$	0.5%
$h_{sat}(l)$	$270 \le T \le 475$	0.03 kJ/kg
$h_{sat}(g)$	$270 \le T \le 475$	0.03 kJ/kg
$\Delta_{vap}h$	$270 \le T \le 450$	0.02%
	$450 < T \le 475$	0.2%
$s_{sat}(l)$	$270 \le T \le 475$	0.0001 kJ/kg K
$s_{sat}(g)$	$270 \le T \le 475$	0.0002 kJ/kg K
$c_{sat}(l)$	$270 \le T \le 470$	0.01 kJ/kg K

Errors arising from linear interpolation between adjacent tabulated values in Table 6.3 at 1 MPa pressure are shown in Table 4.3. In general these are very small. At other pressures, interpolation errors are of the same order of magnitude.

TABLE 4.3
Typical errors due to linear interpolation of the data in Table 6.3

Single phase property	Temperature range K	Maximum error due to linear interpolation
Specific enthalpy of liquid, h(l)	$280 \le T \le 420$	0.07 kJ/kg
Specific enthalpy of vapour, h(g)	$440 \le T \le 470$	0.4 kJ/kg
	$470 < T \le 580$	0.06 kJ/kg
Specific entropy of liquid, s(l)	$280 \le T \le 420$	0.0001 kJ/kg K
Specific entropy of vapour, s(g)	$440 \le T \le 470$	0.0015 kJ/kg K
	$470 < T \le 580$	0.0002 kJ/kg K
Isobaric specific heat capacity, c_p		0.01 kJ/kg K
Isochoric specific heat capacity, c_v		0.01 kJ/kg K

5. REFERENCES AND DERIVATION

References

The references given are recommended sources of information supplementary to that in this Data Item.

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Derivation

The Derivation lists selected sources that have assisted in the preparation of this Data Item.

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6. TABLES

C_3H_8O PROPAN-2-OL (ISOPROPYL ALCOHOL)
CAS Reg. No: 67-63-0

TABLE 6.1 Physical constants of propan-2-ol

Physical Constants	*Symbol*	*Value*	
Molar Gas Constant	R	8.314 3 J/mol K	
Specific Gas Constant	r	138.350 J/kg K	
Molar Mass	M	0.060096 kg/mol	
Critical Temperature	T_c	508.3 K	(235.15 ^oC)
Critical Pressure	p_c	4763. kPa	
Critical Density	ρ_c	273.2 kg/m^3	
Critical Molar Volume	V_c	220. cm^3/mol	
Normal Boiling Temperature	T_b	355.39 K	(82.24 ^oC)
Normal Melting Temperature	T_m	185.2 K	(-87.95 ^oC)
Acentric Factor	ω	0.665	

TABLE 6.2
Thermodynamic properties of propan-2-ol on the saturation line at fixed pressures

p	T_{68}	ρ(l)	v(g)	h(l)	$\Delta_{vap}h$	h(g)	s(l)	s(g)	c_{sat}(l)
kPa	K	kg/m^3	m^3/kg	kJ/kg	kJ/kg	kJ/kg	kJ/kg K	kJ/kg K	kJ/kg K
1	272.75	800.38	37.720	413.33	778.48	1191.8	2.7951	5.6554	2.28
2	282.15	793.63	19.499	435.43	769.67	1205.1	2.8748	5.6075	2.40
3	288.01	789.19	13.262	449.67	763.84	1213.5	2.9248	5.5810	2.48
4	292.35	785.79	10.090	460.48	759.31	1219.8	2.9620	5.5629	2.53
5	295.83	782.99	8.1638	469.31	755.54	1224.9	2.9920	5.5493	2.57
6	298.75	780.60	6.8665	476.83	752.28	1229.1	3.0173	5.5385	2.61
7	301.27	778.49	5.9320	483.43	749.37	1232.8	3.0393	5.5295	2.65
8	303.49	776.59	5.2261	489.33	746.73	1236.1	3.0588	5.5219	2.67
9	305.49	774.88	4.6735	494.68	744.31	1239.0	3.0764	5.5153	2.70
10	307.30	773.30	4.2289	499.59	742.07	1241.7	3.0924	5.5095	2.72
12	310.50	770.46	3.5571	508.36	738.00	1246.4	3.1208	5.4997	2.76
14	313.27	767.97	3.0730	516.07	734.36	1250.4	3.1455	5.4915	2.80
16	315.73	765.73	2.7072	522.98	731.04	1254.0	3.1674	5.4846	2.83
18	317.93	763.70	2.4208	529.26	727.99	1257.2	3.1873	5.4786	2.86
20	319.93	761.83	2.1903	535.03	725.14	1260.2	3.2053	5.4734	2.89
22	321.77	760.10	2.0007	540.37	722.47	1262.8	3.2220	5.4687	2.91
25	324.27	757.72	1.7718	547.74	718.74	1266.5	3.2448	5.4625	2.94
30	327.94	754.20	1.4898	558.66	713.10	1271.8	3.2782	5.4538	2.99
35	331.11	751.11	1.2864	568.28	708.02	1276.3	3.3074	5.4465	3.03
40	333.93	748.34	1.1326	576.92	703.36	1280.3	3.3334	5.4404	3.07
45	336.47	745.82	1.0122	584.79	699.05	1283.8	3.3568	5.4350	3.10
50	338.78	743.50	0.91531	592.03	695.00	1287.0	3.3783	5.4302	3.14
60	342.87	739.33	0.76877	605.04	687.59	1292.6	3.4164	5.4221	3.20
70	346.43	735.62	0.66312	616.53	680.86	1297.4	3.4497	5.4152	3.24
80	349.60	732.27	0.58324	626.87	674.66	1301.5	3.4794	5.4092	3.28
90	352.45	729.19	0.52068	636.30	668.89	1305.2	3.5062	5.4040	3.31
100	355.06	726.33	0.47032	645.00	663.47	1308.5	3.5307	5.3993	3.34
101.325	355.39	725.97	0.46438	646.10	662.78	1308.9	3.5338	5.3987	3.34
150	365.57	714.33	0.31728	680.91	640.10	1321.0	3.6302	5.3808	3.46
200	373.54	704.70	0.23936	708.98	620.77	1329.8	3.7060	5.3673	3.55
250	380.06	696.50	0.19205	732.43	604.00	1336.4	3.7680	5.3566	3.62
300	385.61	689.26	0.16024	752.78	589.06	1341.8	3.8210	5.3479	3.68
350	390.48	682.71	0.13738	770.89	575.54	1346.4	3.8675	5.3406	3.73
400	394.84	676.69	0.12016	787.30	563.17	1350.5	3.9091	5.3345	3.78
500	402.42	665.83	0.09593	816.32	541.16	1357.5	3.9815	5.3252	3.85
600	408.92	656.08	0.07971	841.66	522.02	1363.7	4.0436	5.3190	3.91
700	414.65	647.14	0.06809	864.33	505.12	1369.4	4.0983	5.3151	3.96
800	419.79	638.80	0.05936	884.97	490.01	1375.0	4.1474	5.3131	4.01
900	424.47	630.91	0.05254	903.99	476.36	1380.3	4.1920	5.3126	4.05
1000	428.78	623.39	0.04707	921.69	463.90	1385.6	4.2332	5.3133	4.09
1200	436.53	609.19	0.03882	953.95	441.77	1395.7	4.3070	5.3170	4.15
1500	446.53	589.33	0.03044	996.44	413.32	1409.8	4.4021	5.3254	4.23
2000	460.35	558.45	0.02182	1056.4	372.27	1428.7	4.5325	5.3388	4.32
2500	471.80	528.84	0.01641	1107.2	333.19	1440.4	4.6395	5.3437	
3000	481.65	499.39	0.01266	1151.8	292.24	1444.1	4.7310	5.3362	
3500	490.29	468.92	0.009868	1192.3	247.20	1439.5	4.8122	5.3153	
4000	498.00	435.53	0.007652	1230.9	195.61	1426.5	4.8881	5.2802	
4500	504.93	394.74	0.005566	1270.9	127.60	1398.5	4.9654	5.2178	
4763	508.30	273.2	0.003661	1342.6	0	1342.6	5.1053	5.1053	∞

TABLE 6.3
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 1 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
272.75*	0.001249†	0.000†	413.33†	2.7951†	2.31†	1.61†
272.75*	37.72†	1.000*	1191.8*	5.6554*	1.39*	1.25*
270	0.001246	0.000	406.98	2.7718	2.29	1.54
275	38.03	1.000	1195.0	5.6669	1.40	1.26
280	38.72	1.000	1202.0	5.6923	1.42	1.28
285	39.41	1.000	1209.2	5.7176	1.44	1.30
290	40.10	1.000	1216.4	5.7428	1.46	1.32
295	40.79	1.000	1223.7	5.7679	1.48	1.34
300	41.49	1.000	1231.2	5.7929	1.50	1.36
305	42.18	1.000	1238.7	5.8178	1.52	1.38
310	42.87	1.000	1246.3	5.8426	1.53	1.40
315	43.56	1.000	1254.1	5.8673	1.55	1.41
320	44.25	1.000	1261.9	5.8919	1.57	1.43
325	44.95	1.000	1269.8	5.9164	1.59	1.45
330	45.64	1.000	1277.8	5.9409	1.61	1.47
335	46.33	1.000	1285.9	5.9652	1.63	1.49
340	47.02	1.000	1294.0	5.9894	1.65	1.51
345	47.72	1.000	1302.3	6.0136	1.66	1.53
350	48.41	1.000	1310.7	6.0377	1.68	1.54
355	49.10	1.000	1319.1	6.0617	1.70	1.56
360	49.79	1.000	1327.7	6.0856	1.72	1.58
365	50.48	1.000	1336.3	6.1094	1.74	1.60
370	51.18	1.000	1345.1	6.1332	1.76	1.62
375	51.87	1.000	1353.9	6.1568	1.77	1.63
380	52.56	1.000	1362.8	6.1804	1.79	1.65
385	53.25	1.000	1371.8	6.2040	1.81	1.67
390	53.95	1.000	1380.9	6.2274	1.83	1.69
395	54.64	1.000	1390.1	6.2508	1.84	1.71
400	55.33	1.000	1399.3	6.2741	1.86	1.72
405	56.02	1.000	1408.7	6.2973	1.88	1.74
410	56.71	1.000	1418.1	6.3205	1.90	1.76
415	57.41	1.000	1427.6	6.3436	1.91	1.78
420	58.10	1.000	1437.3	6.3666	1.93	1.79
425	58.79	1.000	1447.0	6.3896	1.95	1.81
430	59.48	1.000	1456.7	6.4125	1.97	1.83
435	60.17	1.000	1466.6	6.4353	1.98	1.84
440	60.87	1.000	1476.6	6.4581	2.00	1.86
445	61.56	1.000	1486.6	6.4808	2.02	1.88
450	62.25	1.000	1496.7	6.5034	2.03	1.89
455	62.94	1.000	1506.9	6.5260	2.05	1.91
460	63.63	1.000	1517.2	6.5485	2.07	1.93
465	64.33	1.000	1527.6	6.5709	2.08	1.94
470	65.02	1.000	1538.1	6.5932	2.10	1.96
475	65.71	1.000	1548.6	6.6155	2.12	1.98
480	66.40	1.000	1559.2	6.6378	2.13	1.99
485	67.09	1.000	1569.9	6.6599	2.15	2.01
490	67.78	1.000	1580.7	6.6820	2.16	2.02
495	68.48	1.000	1591.5	6.7041	2.18	2.04
500	69.17	1.000	1602.5	6.7261	2.19	2.06
505	69.86	1.000	1613.5	6.7480	2.21	2.07
510	70.55	1.000	1624.6	6.7698	2.22	2.09
515	71.24	1.000	1635.7	6.7916	2.24	2.10
520	71.94	1.000	1647.0	6.8133	2.26	2.12
525	72.63	1.000	1658.3	6.8350	2.27	2.13
530	73.32	1.000	1669.7	6.8565	2.28	2.15
535	74.01	1.000	1681.1	6.8781	2.30	2.16
540	74.70	1.000	1692.7	6.8995	2.31	2.18
545	75.40	1.000	1704.3	6.9209	2.33	2.19
550	76.09	1.000	1716.0	6.9423	2.34	2.20
555	76.78	1.000	1727.7	6.9635	2.36	2.22
560	77.47	1.000	1739.5	6.9847	2.37	2.23
565	78.16	1.000	1751.4	7.0059	2.39	2.25
570	78.85	1.000	1763.4	7.0270	2.40	2.26
575	79.55	1.000	1775.4	7.0480	2.41	2.27
580	80.24	1.000	1787.5	7.0689	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 1.5 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
278.16*	0.001255†	0.000†	425.95†	2.8410†	2.35†	1.76†
278.16†	25.64*	0.999*	1199.4*	5.6270*	1.41*	1.27*
270	0.001246	0.000	406.98	2.7718	2.29	1.54
275	0.001252	0.000	418.54	2.8142	2.33	1.67
280	25.81	0.999	1202.0	5.6363	1.42	1.28
285	26.27	0.999	1209.2	5.6616	1.44	1.30
290	26.73	0.999	1216.4	5.6868	1.46	1.32
295	27.19	0.999	1223.7	5.7118	1.48	1.34
300	27.65	0.999	1231.2	5.7368	1.50	1.36
305	28.11	0.999	1238.7	5.7617	1.51	1.38
310	28.57	0.999	1246.3	5.7865	1.53	1.40
315	29.03	0.999	1254.0	5.8112	1.55	1.41
320	29.50	0.999	1261.8	5.8358	1.57	1.43
325	29.96	0.999	1269.7	5.8603	1.59	1.45
330	30.42	0.999	1277.7	5.8847	1.61	1.47
335	30.88	0.999	1285.8	5.9090	1.63	1.49
340	31.34	1.000	1294.0	5.9333	1.65	1.51
345	31.81	1.000	1302.3	5.9574	1.66	1.53
350	32.27	1.000	1310.7	5.9815	1.68	1.54
355	32.73	1.000	1319.1	6.0055	1.70	1.56
360	33.19	1.000	1327.7	6.0294	1.72	1.58
365	33.65	1.000	1336.3	6.0532	1.74	1.60
370	34.11	1.000	1345.0	6.0770	1.76	1.62
375	34.58	1.000	1353.9	6.1007	1.77	1.63
380	35.04	1.000	1362.8	6.1243	1.79	1.65
385	35.50	1.000	1371.8	6.1478	1.81	1.67
390	35.96	1.000	1380.9	6.1713	1.83	1.69
395	36.42	1.000	1390.0	6.1947	1.84	1.71
400	36.88	1.000	1399.3	6.2180	1.86	1.72
405	37.35	1.000	1408.7	6.2412	1.88	1.74
410	37.81	1.000	1418.1	6.2644	1.90	1.76
415	38.27	1.000	1427.6	6.2875	1.91	1.78
420	38.73	1.000	1437.2	6.3105	1.93	1.79
425	39.19	1.000	1446.9	6.3335	1.95	1.81
430	39.65	1.000	1456.7	6.3564	1.97	1.83
435	40.11	1.000	1466.6	6.3792	1.98	1.84
440	40.57	1.000	1476.6	6.4020	2.00	1.86
445	41.04	1.000	1486.6	6.4247	2.02	1.88
450	41.50	1.000	1496.7	6.4473	2.03	1.89
455	41.96	1.000	1506.9	6.4699	2.05	1.91
460	42.42	1.000	1517.2	6.4923	2.07	1.93
465	42.88	1.000	1527.6	6.5148	2.08	1.94
470	43.34	1.000	1538.1	6.5371	2.10	1.96
475	43.80	1.000	1548.6	6.5594	2.12	1.98
480	44.27	1.000	1559.2	6.5817	2.13	1.99
485	44.73	1.000	1569.9	6.6038	2.15	2.01
490	45.19	1.000	1580.7	6.6259	2.16	2.02
495	45.65	1.000	1591.5	6.6480	2.18	2.04
500	46.11	1.000	1602.5	6.6700	2.19	2.06
505	46.57	1.000	1613.5	6.6919	2.21	2.07
510	47.03	1.000	1624.6	6.7137	2.22	2.09
515	47.49	1.000	1635.7	6.7355	2.24	2.10
520	47.96	1.000	1647.0	6.7572	2.26	2.12
525	48.42	1.000	1658.3	6.7789	2.27	2.13
530	48.88	1.000	1669.7	6.8004	2.28	2.15
535	49.34	1.000	1681.1	6.8220	2.30	2.16
540	49.80	1.000	1692.7	6.8434	2.31	2.18
545	50.26	1.000	1704.3	6.8648	2.33	2.19
550	50.72	1.000	1715.9	6.8862	2.34	2.20
555	51.18	1.000	1727.7	6.9074	2.36	2.22
560	51.65	1.000	1739.5	6.9286	2.37	2.23
565	52.11	1.000	1751.4	6.9498	2.39	2.25
570	52.57	1.000	1763.4	6.9709	2.40	2.26
575	53.03	1.000	1775.4	6.9919	2.41	2.27
580	53.49	1.000	1787.5	7.0128	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 2 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
282.15*	0.001260†	0.000†	435.43†	2.8748†	2.40†	1.86†
282.15†	19.50*	0.999*	1205.1*	5.6075*	1.42*	1.29*
270	0.001246	0.000	406.99	2.7718	2.29	1.54
275	0.001252	0.000	418.54	2.8142	2.33	1.67
280	0.001258	0.000	430.30	2.8565	2.38	1.81
285	19.70	0.999	1209.2	5.6219	1.44	1.30
290	20.04	0.999	1216.4	5.6470	1.46	1.32
295	20.39	0.999	1223.7	5.6720	1.48	1.34
300	20.73	0.999	1231.2	5.6970	1.49	1.36
305	21.08	0.999	1238.7	5.7219	1.51	1.38
310	21.42	0.999	1246.3	5.7466	1.53	1.39
315	21.77	0.999	1254.0	5.7713	1.55	1.41
320	22.12	0.999	1261.8	5.7959	1.57	1.43
325	22.46	0.999	1269.7	5.8204	1.59	1.45
330	22.81	0.999	1277.7	5.8448	1.61	1.47
335	23.16	0.999	1285.8	5.8692	1.63	1.49
340	23.50	0.999	1294.0	5.8934	1.65	1.51
345	23.85	0.999	1302.3	5.9176	1.66	1.53
350	24.20	0.999	1310.6	5.9416	1.68	1.54
355	24.54	0.999	1319.1	5.9656	1.70	1.56
360	24.89	0.999	1327.6	5.9896	1.72	1.58
365	25.24	0.999	1336.3	6.0134	1.74	1.60
370	25.58	1.000	1345.0	6.0372	1.76	1.62
375	25.93	1.000	1353.8	6.0608	1.77	1.63
380	26.28	1.000	1362.7	6.0844	1.79	1.65
385	26.62	1.000	1371.7	6.1080	1.81	1.67
390	26.97	1.000	1380.8	6.1314	1.83	1.69
395	27.31	1.000	1390.0	6.1548	1.84	1.71
400	27.66	1.000	1399.3	6.1781	1.86	1.72
405	28.01	1.000	1408.6	6.2014	1.88	1.74
410	28.35	1.000	1418.1	6.2246	1.90	1.76
415	28.70	1.000	1427.6	6.2477	1.91	1.78
420	29.04	1.000	1437.2	6.2707	1.93	1.79
425	29.39	1.000	1446.9	6.2937	1.95	1.81
430	29.74	1.000	1456.7	6.3166	1.97	1.83
435	30.08	1.000	1466.6	6.3394	1.98	1.84
440	30.43	1.000	1476.5	6.3622	2.00	1.86
445	30.78	1.000	1486.6	6.3848	2.02	1.88
450	31.12	1.000	1496.7	6.4075	2.03	1.90
455	31.47	1.000	1506.9	6.4300	2.05	1.91
460	31.81	1.000	1517.2	6.4525	2.07	1.93
465	32.16	1.000	1527.6	6.4750	2.08	1.94
470	32.51	1.000	1538.0	6.4973	2.10	1.96
475	32.85	1.000	1548.6	6.5196	2.12	1.98
480	33.20	1.000	1559.2	6.5419	2.13	1.99
485	33.54	1.000	1569.9	6.5640	2.15	2.01
490	33.89	1.000	1580.7	6.5861	2.16	2.02
495	34.24	1.000	1591.5	6.6082	2.18	2.04
500	34.58	1.000	1602.5	6.6301	2.19	2.06
505	34.93	1.000	1613.5	6.6521	2.21	2.07
510	35.27	1.000	1624.6	6.6739	2.22	2.09
515	35.62	1.000	1635.7	6.6957	2.24	2.10
520	35.97	1.000	1647.0	6.7174	2.26	2.12
525	36.31	1.000	1658.3	6.7390	2.27	2.13
530	36.66	1.000	1669.7	6.7606	2.28	2.15
535	37.00	1.000	1681.1	6.7822	2.30	2.16
540	37.35	1.000	1692.7	6.8036	2.31	2.18
545	37.70	1.000	1704.3	6.8250	2.33	2.19
550	38.04	1.000	1715.9	6.8463	2.34	2.20
555	38.39	1.000	1727.7	6.8676	2.36	2.22
560	38.73	1.000	1739.5	6.8888	2.37	2.23
565	39.08	1.000	1751.4	6.9100	2.39	2.25
570	39.42	1.000	1763.4	6.9310	2.40	2.26
575	39.77	1.000	1775.4	6.9521	2.41	2.27
580	40.12	1.000	1787.5	6.9730	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 3 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
288.01*	0.001267†	0.000†	449.67†	2.9248†	2.46†	2.02†
288.01†	13.26*	0.998*	1213.5*	5.5810*	1.44*	1.31*
270	0.001246	0.000	406.99	2.7718	2.29	1.54
275	0.001252	0.000	418.54	2.8142	2.33	1.67
280	0.001258	0.000	430.30	2.8565	2.37	1.81
285	0.001263	0.000	442.30	2.8990	2.43	1.94
290	13.35	0.998	1216.4	5.5909	1.45	1.31
295	13.58	0.998	1223.7	5.6159	1.47	1.33
300	13.81	0.999	1231.1	5.6409	1.49	1.35
305	14.05	0.999	1238.6	5.6657	1.51	1.37
310	14.28	0.999	1246.2	5.6905	1.53	1.39
315	14.51	0.999	1254.0	5.7151	1.55	1.41
320	14.74	0.999	1261.8	5.7397	1.57	I.43
325	14.97	0.999	1269.7	5.7642	1.59	1.45
330	15.20	0.999	1277.7	5.7886	1.61	1.47
335	15.43	0.999	1285.7	5.8129	1.63	1.49
340	15.66	0.999	1293.9	5.8372	1.65	1.51
345	15.90	0.999	1302.2	5.8'613	1.66	1.53
350	16.13	0.999	1310.6	5.8854	1.68	1.54
355	16.36	0.999	1319.0	5.9094	1.70	1.56
360	16.59	0.999	1327.6	5.9333	1.72	1.58
365	16.82	0.999	1336.2	5.9572	1.74	1.60
370	17.05	0.999	1345.0	5.9810	1.76	1.62
375	17.28	0.999	1353.8	6.0046	1.77	1.63
380	17.51	0.999	1362.7	6.0283	1.79	1.65
385	17.74	0.999	1371.7	6.0518	1.81	1.67
390	17.98	0.999	1380.8	6.0753	1.83	1.69
395	18.21	0.999	1390.0	6.0987	1.85	1.71
400	18.44	0.999	1399.2	6.1220	1.86	1.72
405	18.67	1.000	1408.6	6.1452	1.88	1.74
410	18.90	1.000	1418.0	6.1684	1.90	1.76
415	19.13	1.000	1427.6	6.1915	1.92	1.78
420	19.36	1.000	1437.2	6.2146	1.93	1.79
425	19.59	1.000	1446.9	6.2375	1.95	1.81
430	19.82	1.000	1456.7	6.2604	1.97	1.83
435	20.05	1.000	1466.6	6.2833	1.98	1.84
440	20.28	1.000	1476.5	6.3060	2.00	1.86
445	20.51	1.000	1486.6	6.3287	2.02	1.88
450	20.75	1.000	1496.7	6.3513	2.03	1.90
455	20.98	1.000	1506.9	6.3739	2.05	1.91
460	21.21	1.000	1517.2	6.3964	2.07	1.93
465	21.44	1.000	1527.6	6.4188	2.08	1.94
470	21.67	1.000	1538.0	6.4412	2.10	1.96
475	21.90	1.000	1540.6	6.4635	2.12	1.98
480	22.13	1.000	1559.2	6.4857	2.13	1.99
485	22.36	1.000	1569.9	6.5079	2.15	2.01
490	22.59	1.000	1580.7	6.5300	2.16	2.02
495	22.82	1.000	1591.5	6.5521	2.18	2.04
500	23.05	1.000	1602.4	6.5740	2.19	2.06
505	23.28	1.000	1613.5	6.5959	2.21	2.07
510	23.51	1.000	1624.5	6.6178	2.22	2.09
515	23.74	1.000	1635.7	6.6396	2.24	2.10
520	23.98	1.000	1646.9	6.6613	2.26	2.12
525	24.21	1.000	1658.3	6.6829	2.27	2.13
530	24.44	1.000	1669.6	6.7045	2.28	2.15
535	24.67	1.000	1681.1	6.7260	2.30	2.16
540	24.90	1.000	1692.6	6.7475	2.31	2.18
545	25.13	1.000	1704.2	6.7689	2.33	2.19
550	25.36	1.000	1715.9	6.7902	2.34	2.20
555	25.59	1.000	1727.7	6.8115	2.36	2.22
560	25.82	1.000	1739.5	6.8327	2.37	2.23
565	26.05	1.000	1751.4	6.8538	2.39	2.25
570	26.28	1.000	1763.3	6.8749	2.40	2.26
575	26.51	1.000	1775.4	6.8959	2.41	2.27
580	26.74	1.000	1787.5	6.9169	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 4 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
292.35*	0.001273†	0.000†	460.48†	2.9620†	2.52†	2.13†
292.35†	10.09*	0.998*	1219.8*	5.5629*	1.46*	1.32*
270	0.001246	0.000	406.99	2.7718	2.29	1.54
275	0.001252	0.000	418.55	2.8142	2.33	1.67
280	0.001258	0.000	430.30	2.8565	2.37	1.81
285	0.001263	0.000	442.30	2.8990	2.43	1.94
290	0.001270	0.000	454.59	2.9418	2.49	2.07
295	10.18	0.998	1223.7	5.5761	1.47	1.33
300	10.36	0.998	1231.1	5.6010	1.49	1.35
305	10.53	0.998	1238.6	5.6258	1.51	1.37
310	10.70	0.998	1246.2	5.6505	1.53	1.39
315	10.88	0.998	1253.9	5.6752	1.55	1.41
320	11.05	0.998	1261.7	5.6998	1.57	1.43
325	11.22	0.998	1269.6	5.7243	1.59	1.45
330	11.40	0.998	1277.6	5.7487	1.61	1.47
335	11.57	0.999	1285.7	5.7730	1.63	1.49
340	11.74	0.999	1293.9	5.7972	1.65	1.51
345	11.92	0.999	1302.1	5.8214	1.66	1.53
350	12.09	0.999	1310.5	5.8455	1.68	1.54
355	12.26	0.999	1319.0	5.8695	1.70	1.56
360	12.44	0.999	1327.5	5.8934	1.72	1.58
365	12.61	0.999	1336.2	5.9173	1.74	1.60
370	12.79	0.999	1344.9	5.9411	1.76	1.62
375	12.96	0.999	1353.7	5.9647	1.77	1.64
380	13.13	0.999	1362.6	5.9884	1.79	1.65
385	13.31	0.999	1371.7	6.0119	1.81	1.67
390	13.48	0.999	1380.7	6.0354	1.83	1.69
395	13.65	0.999	1389.9	6.0588	1.85	1.71
400	13.83	0.999	1399.2	6.0821	1.86	1.72
405	14.00	0.999	1408.6	6.1054	1.88	1.74
410	14.17	0.999	1418.0	6.1285	1.90	1.76
415	14.35	0.999	1427.5	6.1517	1.92	1.78
420	14.52	0.999	1437.2	6.1747	1.93	1.79
425	14.69	0.999	1446.9	6.1977	1.95	1.81
430	14.86	0.999	1456.7	6.2206	1.97	1.83
435	15.04	0.999	1466.5	6.2434	1.98	1.85
440	15.21	1.000	1476.5	6.2662	2.00	1.86
445	15.38	1.000	1486.5	6.2889	2.02	1.88
450	15.56	1.000	1496.7	6.3115	2.03	1.90
455	15.73	1.000	1506.9	6.3341	2.05	1.91
460	15.90	1.000	1517.2	6.3566	2.07	1.93
465	16.08	1.000	1527.6	6.3790	2.08	1.94
470	16.25	1.000	1538.0	6.4014	2.10	1.96
475	16.42	1.000	1548.5	6.4237	2.12	1.98
480	16.60	1.000	1559.2	6.4459	2.13	1.99
485	16.77	1.000	1569.9	6.4681	2.15	2.01
490	16.94	1.000	1580.6	6.4902	2.16	2.02
495	17.11	1.000	1591.5	6.5122	2.18	2.04
500	17.29	1.000	1602.4	6.5342	2.19	2.06
505	17.46	1.000	1613.4	6.5561	2.21	2.07
510	17.63	1.000	1624.5	6.5780	2.22	2.09
515	17.81	1.000	1635.7	6.5997	2.24	2.10
520	17.98	1.000	1646.9	6.6215	2.26	2.12
525	18.15	1.000	1658.2	6.6431	2.27	2.13
530	18.33	1.000	1669.6	6.6647	2.28	2.15
535	18.50	1.000	1681.1	6.6862	2.30	2.16
540	18.67	1.000	1692.6	6.7077	2.31	2.18
545	18.85	1.000	1704.2	6.7291	2.33	2.19
550	19.02	1.000	1715.9	6.7504	2.34	2.20
555	19.19	1.000	1727.7	6.7717	2.36	2.22
560	19.36	1.000	1739.5	6.7929	2.37	2.23
565	19.54	1.000	1751.4	6.8140	2.39	2.25
570	19.71	1.000	1763.3	6.8351	2.40	2.26
575	19.88	1.000	1775.4	6.8561	2.41	2.27
580	20.06	1.000	1787.5	6.8771	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 5 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
295.83*	0.001277†	0.000†	469.31†	2.9920†	2.56†	2.22†
295.83†	8.164*	0.997*	1224.9*	5.5493*	1.47*	1.33*
270	0.001246	0.000	406.99	2.7718	2.29	1.54
275	0.001252	0.000	418.55	2.8142	2.33	1.67
280	0.001258	0.000	430.30	2.8565	2.37	1.81
285	0.001263	0.000	442.31	2.8990	2.43	1.94
290	0.001270	0.000	454.59	2.9418	2.49	2.07
295	0.001276	0.000	467.18	2.9848	2.55	2.20
300	8.280	0.997	1231.0	5.5700	1.49	1.35
305	8.418	0.998	1238.5	5.5948	1.51	1.37
310	8.557	0.998	1246.1	5.6195	1.53	1.39
315	8.696	0.998	1253.8	5.6442	1.55	1.41
320	8.836	0.998	1261.6	5.6688	1.57	1.43
325	8.975	0.998	1269.5	5.6933	1.59	1.45
330	9.114	0.998	1277.5	5.7177	1.61	1.47
335	9.253	0.998	1285.6	5.7420	1.63	1.49
340	9.392	0.998	1293.8	5.7662	1.65	1.51
345	9.531	0.998	1302.1	5.7904	1.66	1.53
350	9.670	0.998	1310.4	5.8145	1.68	1.54
355	9.809	0.999	1318.9	5.8385	1.70	1.56
360	9.948	0.999	1327.5	5.8624	1.72	1.58
365	10.09	0.999	1336.1	5.8863	1.74	1.60
370	10.23	0.999	1344.9	5.9101	1.76	1.62
375	10.36	0.999	1353.7	5.9338	1.77	1.64
380	10.50	0.999	1362.6	5.9574	1.79	1.65
385	10.64	0.999	1371.6	5.9810	1.81	1.67
390	10.78	0.999	1380.7	6.0044	1.83	1.69
395	10.92	0.999	1389.9	6.0278	1.85	1.71
400	11.06	0.999	1399.2	6.0512	1.86	1.72
405	11.20	0.999	1408.5	6.0744	1.88	1.74
410	11.34	0.999	1418.0	6.0976	1.90	1.76
415	11.47	0.999	1427.5	6.1207	1.92	1.78
420	11.61	0.999	1437.1	6.1438	1.93	1.79
425	11.75	0.999	1446.8	6.1668	1.95	1.81
430	11.89	0.999	1456.6	6.1897	1.97	1.83
435	12.03	0.999	1466.5	6.2125	1.98	1.85
440	12.17	0.999	1476.5	6.2353	2.00	1.86
445	12.31	0.999	1486.5	6.2580	2.02	1.88
450	12.44	0.999	1496.7	6.2806	2.03	1.90
455	12.58	0.999	1506.9	6.3032	2.05	1.91
460	12.72	0.999	1517.2	6.3257	2.07	1.93
465	12.86	0.999	1527.5	6.3481	2.08	1.94
470	13.00	1.000	1538.0	6.3705	2.10	1.96
475	13.14	1.000	1548.5	6.3928	2.12	1.98
480	13.28	1.000	1559.1	6.4150	2.13	1.99
485	13.41	1.000	1569.8	6.4372	2.15	2.01
490	13.55	1.000	1580.6	6.4593	2.16	2.02
495	13.69	1.000	1591.5	6.4813	2.18	2.04
500	13.83	1.000	1602.4	6.5033	2.19	2.06
505	13.97	1.000	1613.4	6.5252	2.21	2.07
510	14.11	1.000	1624.5	6.5471	2.22	2.09
515	14.24	1.000	1635.7	6.5689	2.24	2.10
520	14.38	1.000	1646.9	6.5906	2.26	2.12
525	14.52	1.000	1658.2	6.6122	2.27	2.13
530	14.66	1.000	1669.6	6.6338	2.28	2.15
535	14.80	1.000	1681.1	6.6553	2.30	2.16
540	14.94	1.000	1692.6	6.6768	2.31	2.18
545	15.08	1.000	1704.2	6.6982	2.33	2.19
550	15.21	1.000	1715.9	6.7195	2.34	2.20
555	15.35	1.000	1727.6	6.7408	2.36	2.22
560	15.49	1.000	1739.5	6.7620	2.37	2.23
565	15.63	1.000	1751.4	6.7831	2.39	2.25
570	15.77	1.000	1763.3	6.8042	2.40	2.26
575	15.91	1.000	1775.3	6.8252	2.41	2.27
580	16.04	1.000	1787.4	6.8462	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 7 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
301.27*	0.001285†	0.000†	483.43†	3.0393†	2.64†	2.35†
301.27†	5.932*	0.996*	1232.8*	5.5295*	1.49*	1.35*
270	0.001246	0.000	406.99	2.7718	2.30	1.54
275	0.001252	0.000	418.55	2.8142	2.33	1.67
280	0.001258	0.000	430.30	2.8565	2.38	1.81
285	0.001263	0.000	442.31	2.8990	2.43	1.94
290	0.001270	0.000	454.59	2.9417	2.49	2.07
295	0.001276	0.000	467.19	2.9848	2.55	2.20
300	0.001283	0.000	480.11	3.0282	2.62	2.32
305	6.006	0.996	1238.4	5.5480	1.51	1.37
310	6.106	0.997	1246.0	5.5727	1.53	1.39
315	6.206	0.997	1253.7	5.5973	1.55	1.41
320	6.305	0.997	1261.5	5.6219	1.57	1.43
325	6.405	0.997	1269.4	5.6464	1.59	1.45
330	6.504	0.997	1277.4	5.6708	1.61	1.47
335	6.604	0.997	1285.5	5.6951	1.63	1.49
340	6.704	0.998	1293.7	5.7194	1.65	1.51
345	6.803	0.998	1301.9	5.7436	1.67	1.53
350	6.903	0.998	1310.3	5.7677	1.68	1.54
355	7.002	0.998	1318.8	5.7917	1.70	1.56
360	7.102	0.998	1327.4	5.8156	1.72	1.58
365	7.201	0.998	1336.0	5.8395	1.74	1.60
370	7.300	0.998	1344.7	5.8633	1.76	1.62
375	7.400	0.998	1353.6	5.8870	1.78	1.64
380	7.499	0.999	1362.5	5.9107	1.79	1.65
385	7.598	0.999	1371.5	5.9342	1.81	1.67
390	7.698	0.999	1380.6	5.9577	1.83	1.69
395	7.797	0.999	1389.8	5.9811	1.85	1.71
400	7.896	0.999	1399.1	6.0045	1.86	1.73
405	7.995	0.999	1408.5	6.0277	1.88	1.74
410	8.094	0.999	1417.9	6.0509	1.90	1.76
415	8.193	0.999	1427.4	6.0741	1.92	1.78
420	8.293	0.999	1437.1	6.0971	1.93	1.79
425	8.392	0.999	1446.8	6.1201	1.95	1.81
430	8.491	0.999	1456.6	6.1430	1.97	1.83
435	8.590	0.999	1466.5	6.1659	1.98	1.85
440	8.689	0.999	1476.4	6.1886	2.00	1.86
445	8.788	0.999	1486.5	6.2114	2.02	1.88
450	8.887	0.999	1496.6	6.2340	2.03	1.90
455	8.986	0.999	1506.8	6.2566	2.05	1.91
460	9.085	0.999	1517.1	6.2791	2.07	1.93
465	9.184	0.999	1527.5	6.3015	2.08	1.94
470	9.283	0.999	1538.0	6.3239	2.10	1.96
475	9.382	0.999	1548.5	6.3462	2.12	1.98
480	9.481	0.999	1559.1	6.3684	2.13	1.99
485	9.579	0.999	1569.8	6.3906	2.15	2.01
490	9.678	0.999	1580.6	6.4127	2.16	2.02
495	9.777	0.999	1591.4	6.4348	2.18	2.04
500	9.876	0.999	1602.4	6.4567	2.19	2.06
505	9.975	0.999	1613.4	6.4786	2.21	2.07
510	10.07	0.999	1624.5	6.5005	2.23	2.09
515	10.17	0.999	1635.6	6.5223	2.24	2.10
520	10.27	0.999	1646.9	6.5440	2.26	2.12
525	10.37	0.999	1658.2	6.5656	2.27	2.13
530	10.47	0.999	1669.6	6.5872	2.28	2.15
535	10.57	1.000	1681.0	6.6087	2.30	2.16
540	10.67	1.000	1692.6	6.6302	2.31	2.18
545	10.77	1.000	1704.2	6.6516	2.33	2.19
550	10.87	1.000	1715.9	6.6729	2.34	2.20
555	10.96	1.000	1727.6	6.6942	2.36	2.22
560	11.06	1.000	1739.4	6.7154	2.37	2.23
565	11.16	1.000	1751.3	6.7365	2.39	2.25
570	11.26	1.000	1763.3	6.7576	2.40	2.26
575	11.36	1.000	1775.3	6.7786	2.41	2.27
580	11.46	1.000	1787.4	6.7996	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 10 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
307.30*	0.001293†	0.000†	499.59†	3.0924†	2.72†	2.49†
307.30†	4.229*	0.995*	1241.7*	5.5095*	1.52*	1.38*
270	0.001246	0.000	406.99	2.7718	2.30	1.54
275	0.001252	0.000	418.55	2.8142	2.33	1.67
280	0.001258	0.000	430.31	2.8565	2.37	1.81
285	0.001263	0.000	442.31	2.8990	2.43	1.94
290	0.001270	0.000	454.60	2.9417	2.49	2.07
295	0.001276	0.000	467.19	2.9848	2.55	2.20
300	0.001283	0.000	480.11	3.0282	2.62	2.32
305	0.001290	0.000	493.37	3.0721	2.69	2.44
310	4.267	0.995	1245.8	5.5228	1.53	1.39
315	4.337	0.995	1253.5	5.5475	1.55	1.41
320	4.407	0.995	1261.3	5.5720	1.57	1.43
325	4.477	0.996	1269.2	5.5965	I.59	1.45
330	4.547	0.996	1277.2	5.6210	1.61	1.47
335	4.617	0.996	1285.3	5.6453	1.63	1.49
340	4.687	0.996	1293.5	5.6696	1.65	1.51
345	4.757	0.997	1301.7	5.6938	1.67	1.53
350	4.827	0.997	1310.1	5.7179	1.69	1.54
355	4.897	0.997	1318.6	5.7420	1.70	1.56
360	4.967	0.997	1327.2	5.7659	1.72	1.58
365	5.037	0.997	1335.8	5.7898	1.74	1.60
370	5.107	0.998	1344.6	5.8136	1.76	1.62
375	5.176	0.998	1353.4	5.8374	1.78	1.64
380	5.246	0.998	1362.4	5.8610	1.80	1.65
385	5.316	0.998	1371.4	5.8846	1.81	1.67
390	5.385	0.998	1380.5	5.9081	1.83	1.69
395	5.455	0.998	1389.7	5.9315	1.85	1.71
400	5.524	0.998	1399.0	5.9549	1.87	1.73
405	5.594	0.998	1408.3	5.9782	1.88	1.74
410	5.663	0.998	1417.8	6.0014	1.90	1.76
415	5.733	0.998	1427.4	6.0245	1.92	1.78
420	5.802	0.999	1437.0	6.0476	1.93	1.80
425	5.872	0.999	1446.7	6.0706	1.95	1.81
430	5.941	0.999	1456.5	6.0935	1.97	1.83
435	6.011	0.999	1466.4	6.1164	1.99	1.85
440	6.080	0.999	1476.4	6.1392	2.00	1.86
445	6.149	0.999	1486.4	6.1619	2.02	1.88
450	6.219	0.999	1496.5	6.1845	2.04	1.90
455	6.288	0.999	1506.8	6.2071	2.05	1.91
460	6.357	0.999	1517.1	6.2296	2.07	1.93
465	6.427	0.999	1527.4	6.2521	2.08	1.95
470	6.496	0.999	1537.9	6.2745	2.10	1.96
475	6.565	0.999	1548.4	6.2968	2.12	1.98
480	6.634	0.999	1559.1	6.3190	2.13	1.99
485	6.704	0.999	1569.8	6.3412	2.15	2.01
490	6.773	0.999	1580.5	6.3633	2.16	2.02
495	6.842	0.999	1591.4	6.3853	2.18	2.04
500	6.912	0.999	1602.3	6.4073	2.19	2.06
505	6.981	0.999	1613.3	6.4292	2.21	2.07
510	7.050	0.999	1624.4	6.4511	2.23	2.09
515	7.119	0.999	1635.6	6.4729	2.24	2.10
520	7.189	0.999	1646.8	6.4946	2.26	2.12
525	7.258	0.999	1658.1	6.5162	2.27	2.13
530	7.327	0.999	1669.5	6.5378	2.28	2.15
535	7.396	0.999	1681.0	6.5593	2.30	2.16
540	7.466	0.999	1692.5	6.5808	2.31	2.18
545	7.535	0.999	1704.1	6.6022	2.33	2.19
550	7.604	0.999	1715.8	6.6235	2.34	2.20
555	7.674	0.999	1727.6	6.6448	2.36	2.22
560	7.743	0.999	1739.4	6.6660	2.37	2.23
565	7.812	0.999	1751.3	6.6871	2.38	2.25
570	7.881	0.999	1763.2	6.7082	2.40	2.26
575	7.951	0.999	1775.3	6.7292	2.41	2.27
580	8.020	0.999	1787.4	6.7502	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 15 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
314.53*	0.001304†	0.000†	519.62†	3.1568†	2.82†	2.64†
314.53†	2.878*	0.992*	1252.3*	5.4879*	1.55*	1.40*
270	0.001246	0.001	406.99	2.7718	2.30	1.54
275	0.001252	0.000	418.56	2.8141	2.33	1.67
280	0.001258	0.000	430.31	2.8565	2.37	1.81
285	0.001263	0.000	442.31	2.8990	2.43	1.94
290	0.001270	0.000	454.60	2.9417	2.49	2.07
295	0.001276	0.000	467.19	2.9848	2.55	2.20
300	0.001283	0.000	480.11	3.0282	2.62	2.32
305	0.001290	0.000	493.37	3.0721	2.69	2.44
310	0.001297	0.000	506.98	3.1163	2.76	2.55
315	2.883	0.992	1253.0	5.4902	1.55	1.41
320	2.930	0.993	1260.8	5.5149	1.57	1.43
325	2.977	0.993	1268.7	5.5394	1.59	1.45
330	3.025	0.994	1276.8	5.5639	1.61	1.47
335	3.072	0.994	1284.9	5.5883	1.63	1.49
340	3.119	0.995	1293.1	5.6126	1.65	1.51
345	3.166	0.995	1301.4	5.6369	1.67	1.53
350	3.213	0.995	1309.8	5.6610	1.69	1.55
355	3.260	0.996	1318.3	5.6851	1.71	1.57
360	3.307	0.996	1326.9	5.7091	1.73	1.58
365	3.353	0.996	1335.5	5.7331	1.74	1.60
370	3.400	0.996	1344.3	5.7569	1.76	1.62
375	3.447	0.997	1353.2	5.7807	1.78	1.64
380	3.493	0.997	1362.1	5.8044	1.80	1.66
385	3.540	0.997	1371.1	5.8280	1.82	1.67
390	3.587	0.997	1380.3	5.8516	1.83	1.69
395	3.633	0.997	1389.5	5.8751	1.85	1.71
400	3.680	0.997	1398.8	5.8984	1.87	1.73
405	3.726	0.998	1408.2	5.9218	1.89	1.74
410	3.773	0.998	1417.6	5.9450	1.90	1.76
415	3.819	0.998	1427.2	5.9682	1.92	1.78
420	3.865	0.998	1436.8	5.9913	1.94	1.80
425	3.912	0.998	1446.6	6.0143	1.95	1.81
430	3.958	0.998	1456.4	6.0372	1.97	1.83
435	4.004	0.998	1466.3	6.0601	1.99	1.85
440	4.051	0.998	1476.2	6.0829	2.00	1.86
445	4.097	0.998	1486.3	6.1056	2.02	1.88
450	4.143	0.998	1496.4	6.1283	2.04	1.90
455	4.190	0.998	1506.7	6.1509	2.05	1.91
460	4.236	0.998	1517.0	6.1734	2.07	1.93
465	4.282	0.998	1527.3	6.1959	2.08	1.95
470	4.328	0.999	1537.8	6.2182	2.10	1.96
475	4.375	0.999	1548.4	6.2406	2.12	1.98
480	4.421	0.999	1559.0	6.2628	2.13	1.99
485	4.467	0.999	1569.7	6.2850	2.15	2.01
490	4.513	0.999	1580.5	6.3071	2.16	2.02
495	4.560	0.999	1591.3	6.3292	2.18	2.04
500	4.606	0.999	1602.3	6.3511	2.19	2.06
505	4.652	0.999	1613.3	6.3730	2.21	2.07
510	4.698	0.999	1624.4	6.3949	2.23	2.09
515	4.744	0.999	1635.5	6.4167	2.24	2.10
520	4.791	0.999	1646.8	6.4384	2.26	2.12
525	4.837	0.999	1658.1	6.4600	2.27	2.13
530	4.883	0.999	1669.5	6.4816	2.28	2.15
535	4.929	0.999	1680.9	6.5032	2.30	2.16
540	4.975	0.999	1692.5	6.5246	2.31	2.17
545	5.022	0.999	1704.1	6.5460	2.33	2.19
550	5.068	0.999	1715.7	6.5673	2.34	2.20
555	5.114	0.999	1727.5	6.5886	2.36	2.22
560	5.160	0.999	1739.3	6.6098	2.37	2.23
565	5.207	0.999	1751.2	6.6309	2.38	2.25
570	5.253	0.999	1763.1	6.6520	2.40	2.26
575	5.299	0.999	1775.2	6.6730	2.41	2.27
580	5.345	0.999	1787.3	6.6939	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 20 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
319.93*	0.001313†	0.001†	535.03†	3.2053†	2.89†	2.74†
319.93†	2.190*	0.990*	1260.2*	5.4734*	1.58*	1.43*
270	0.001246	0.001	407.01	2.7717	2.30	1.54
275	0.001252	0.001	418.56	2.8141	2.33	1.67
280	0.001258	0.001	430.32	2.8565	2.38	1.81
285	0.001263	0.001	442.32	2.8990	2.43	1.94
290	0.001270	0.001	454.60	2.9417	2.49	2.07
295	0.001276	0.001	467.20	2.9848	2.55	2.20
300	0.001283	0.001	480.12	3.0282	2.62	2.32
305	0.001290	0.001	493.38	3.0720	2.69	2.44
310	0.001297	0.001	506.98	3.1163	2.76	2.55
315	0.001305	0.001	520.93	3.1609	2.83	2.65
320	2.191	0.990	1260.3	5.4737	1.58	1.43
325	2.227	0.991	1268.2	5.4984	1.60	1.45
330	2.263	0.991	1276.3	5.5229	1.62	1.47
335	2.298	0.992	1284.4	5.5474	1.64	1.49
340	2.334	0.992	1292.7	5.5718	1.66	1.51
345	2.370	0.993	1301.0	5.5961	1.68	1.53
350	2.405	0.993	1309.4	5.6204	1.69	1.55
355	2.441	0.994	1317.9	5.6445	1.71	1.57
360	2.476	0.994	1326.5	5.6686	1.73	1.59
365	2.511	0.995	1335.2	5.6926	1.75	1.60
370	2.547	0.995	1344.0	5.7165	1.77	1.62
375	2.582	0.995	1352.9	5.7403	1.78	1.64
380	2.617	0.996	1361.9	5.7641	1.80	1.66
385	2.652	0.996	1370.9	5.7878	1.82	1.68
390	2.687	0.996	1380.0	5.8113	1.84	1.69
395	2.722	0.996	1389.3	5.8348	1.85	1.71
400	2.757	0.996	1398.6	5.8583	1.87	1.73
405	2.792	0.997	1408.0	5.8816	1.89	1.75
410	2.827	0.997	1417.5	5.9049	1.90	1.76
415	2.862	0.997	1427.0	5.9281	1.92	1.78
420	2.897	0.997	1436.7	5.9512	1.94	1.80
425	2.932	0.997	1446.4	5.9742	1.96	1.81
430	2.967	0.997	1456.2	5.9972	1.97	1.83
435	3.001	0.997	1466.1	6.0201	1.99	1.85
440	3.036	0.998	1476.1	6.0429	2.00	1.86
445	3.071	0.998	1486.2	6.0656	2.02	1.88
450	3.106	0.998	1496.3	6.0883	2.04	1.90
455	3.141	0.998	1506.5	6.1109	2.05	1.91
460	3.175	0.998	1516.9	6.1335	2.07	1.93
465	3.210	0.998	1527.2	6.1559	2.09	1.95
470	3.245	0.998	1537.7	6.1783	2.10	1.96
475	3.279	0.998	1548.3	6.2006	2.12	1.98
480	3.314	0.998	1558.9	6.2229	2.13	1.99
485	3.349	0.998	1569.6	6.2451	2.15	2.01
490	3.383	0.998	1580.4	6.2672	2.16	2.03
495	3.418	0.998	1591.2	6.2892	2.18	2.04
500	3.453	0.998	1602.2	6.3112	2.20	2.06
505	3.488	0.998	1613.2	6.3332	2.21	2.07
510	3.522	0.998	1624.3	6.3550	2.23	2.09
515	3.557	0.998	1635.4	6.3768	2.24	2.10
520	3.592	0.998	1646.7	6.3985	2.26	2.12
525	3.626	0.999	1658.0	6.4202	2.27	2.13
530	3.661	0.999	1669.4	6.4417	2.28	2.15
535	3.696	0.999	1680.8	6.4633	2.30	2.16
540	3.730	0.999	1692.4	6.4847	2.31	2.17
545	3.765	0.999	1704.0	6.5061	2.33	2.19
550	3.800	0.999	1715.7	6.5274	2.34	2.20
555	3.834	0.999	1727.4	6.5487	2.36	2.22
560	3.869	0.999	1739.2	6.5699	2.37	2.23
565	3.904	0.999	1751.1	6.5910	2.38	2.25
570	3.938	0.999	1763.1	6.6121	2.40	2.26
575	3.973	0.999	1775.1	6.6331	2.41	2.27
580	4.008	0.999	1787.2	6.6540	2.43	2.29

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 30 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
327.94*	0.001326†	0.001†	558.66†	3.2782†	3.00†	2.88†
327.94†	1.490*	0.985*	1271.8*	5.4538*	1.63*	1.47*
270	0.001246	0.001	407.01	2.7717	2.30	1.54
275	0.001252	0.001	418.57	2.8141	2.33	1.67
280	0.001258	0.001	430.32	2.8565	2.37	1.81
285	0.001263	0.001	442.33	2.8990	2.43	1.94
290	0.001270	0.001	454.61	2.9417	2.49	2.07
295	0.001276	0.001	467.21	2.9848	2.55	2.20
300	0.001283	0.001	480.13	3.0282	2.62	2.32
305	0.001290	0.001	493.38	3.0720	2.69	2.44
310	0.001297	0.001	506.99	3.1163	2.76	2.55
315	0.001305	0.001	520.94	3.1609	2.83	2.65
320	0.001313	0.001	535.24	3.2060	2.90	2.74
325	0.001321	0.001	549.89	3.2514	2.96	2.83
330	1.500	0.986	1275.1	5.4640	1.64	1.48
335	1.525	0.987	1283.4	5.4887	1.65	1.50
340	1.549	0.988	1291.7	5.5134	1.67	1.52
345	1.573	0.989	1300.1	5.5379	1.69	1.54
350	1.597	0.990	1308.6	5.5623	1.71	1.56
355	1.621	0.990	1317.1	5.5866	1.72	1.57
360	1.645	0.991	1325.8	5.6109	1.74	1.59
365	1.669	0.992	1334.5	5.6350	1.76	1.61
370	1.693	0.992	1343.4	5.6590	1.78	1.63
375	1.717	0.993	1352.3	5.6830	1.79	1.65
380	1.741	0.993	1361.3	5.7068	1.81	1.66
385	1.764	0.994	1370.4	5.7306	1.83	1.68
390	1.788	0.994	1379.6	5.7543	1.84	1.70
395	1.811	0.994	1388.8	5.7779	1.86	1.72
400	1.835	0.995	1398.2	5.8014	1.88	1.73
405	1.858	0.995	1407.6	5.8248	1.89	1.75
410	1.882	0.995	1417.1	5.8481	1.91	1.77
415	1.905	0.995	1426.7	5.8714	1.93	1.78
420	1.928	0.996	1436.4	5.8945	1.94	1.80
425	1.952	0.996	1446.1	5.9176	1.96	1.82
430	1.975	0.996	1455.9	5.9406	1.98	1.83
435	1.998	0.996	1465.9	5.9636	1.99	1.85
440	2.022	0.996	1475.9	5.9864	2.01	1.87
445	2.045	0.996	1485.9	6.0092	2.02	1.88
450	2.068	0.997	1496.1	6.0319	2.04	1.90
455	2.091	0.997	1506.3	6.0545	2.06	1.92
460	2.115	0.997	1516.7	6.0771	2.07	1.93
465	2.138	0.997	1527.1	6.0996	2.09	1.95
470	2.161	0.997	1537.5	6.1220	2.10	1.96
475	2.184	0.997	1548.1	6.1443	2.12	1.98
480	2.207	0.997	1558.7	6.1666	2.13	1.99
485	2.230	0.997	1569.4	6.1888	2.15	2.01
490	2.254	0.997	1580.2	6.2109	2.17	2.03
495	2.277	0.997	1591.1	6.2330	2.18	2.04
500	2.300	0.997	1602.0	6.2549	2.20	2.06
505	2.323	0.998	1613.0	6.2769	2.21	2.07
510	2.346	0.998	1624.1	6.2987	2.23	2.09
515	2.369	0.998	1635.3	6.3205	2.24	2.10
520	2.393	0.998	1646.5	6.3422	2.26	2.12
525	2.416	0.998	1657.8	6.3639	2.27	2.13
530	2.439	0.998	1669.2	6.3855	2.28	2.15
535	2.462	0.998	1680.7	6.4070	2.30	2.16
540	2.485	0.998	1692.2	6.4284	2.31	2.17
545	2.508	0.998	1703.8	6.4498	2.33	2.19
550	2.531	0.998	1715.5	6.4711	2.34	2.20
555	2.555	0.998	1727.2	6.4924	2.36	2.22
560	2.578	0.998	1739.1	6.5136	2.37	2.23
565	2.601	0.998	1750.9	6.5347	2.38	2.24
570	2.624	0.998	1762.9	6.5558	2.40	2.26
575	2.647	0.998	1774.9	6.5768	2.41	2.27
580	2.670	0.998	1787.0	6.5977	2.42	2.28

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 40 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
333.93*	0.001336†	0.001†	576.92†	3.3334†	3.09†	2.97†
333.93†	1.133*	0.981*	1280.3*	5.4404*	1.67*	1.51*
270	0.001246	0.001	407.02	2.7717	2.30	1.54
275	0.001252	0.001	418.58	2.8141	2.33	1.67
280	0.001258	0.001	430.33	2.8565	2.37	1.81
285	0.001263	0.001	442.34	2.8990	2.43	1.94
290	0.001270	0.001	454.62	2.9417	2.49	2.07
295	0.001276	0.001	467.22	2.9848	2.55	2.20
300	0.001283	0.001	480.14	3.0282	2.62	2.32
305	0.001290	0.001	493.39	3.0720	2.69	2.44
310	0.001297	0.001	507.00	3.1163	2.76	2.55
315	0.001305	0.001	520.95	3.1609	2.83	2.65
320	0.001313	0.001	535.25	3.2059	2.90	2.74
325	0.001321	0.001	549.90	3.2514	2.96	2.83
330	0.001329	0.001	564.89	3.2971	3.03	2.91
335	1.137	0.981	1282.1	5.4457	1.68	1.51
340	1.156	0.983	1290.5	5.4707	1.69	1.53
345	1.174	0.984	1299.0	5.4955	1.71	1.55
350	1.193	0.986	1307.6	5.5202	1.72	1.57
355	1.212	0.987	1316.2	5.5448	1.74	1.58
360	1.230	0.988	1325.0	5.5692	1.75	1.60
365	1.248	0.989	1333.8	5.5935	1.77	1.62
370	1.266	0.989	1342.7	5.6177	1.79	1.64
375	1.284	0.990	1351.7	5.6418	1.80	1.65
380	1.302	0.991	1360.7	5.6658	1.82	1.67
385	1.320	0.991	1369.8	5.6897	1.83	1.69
390	1.338	0.992	1379.1	5.7135	1.85	1.70
395	1.356	0.992	1388.3	5.7371	1.87	1.72
400	1.374	0.993	1397.7	5.7607	1.88	1.74
405	1.391	0.993	1407.2	5.7842	1.90	1.75
410	1.409	0.994	1416.7	5.8076	1.92	1.77
415	1.427	0.994	1426.3	5.8309	1.93	1.79
420	1.444	0.994	1436.0	5.8541	1.95	1.80
425	1.462	0.994	1445.8	5.8773	1.96	1.82
430	1.479	0.995	1455.7	5.9003	1.98	1.84
435	1.497	0.995	1465.6	5.9233	1.99	1.85
440	1.514	0.995	1475.6	5.9462	2.01	1.87
445	1.532	0.995	1485.7	5.9690	2.03	1.89
450	1.549	0.995	1495.9	5.9917	2.04	1.90
455	1.567	0.996	1506.1	6.0144	2.06	1.92
460	1.584	0.996	1516.5	6.0370	2.07	1.93
465	1.602	0.996	1526.9	6.0595	2.09	1.95
470	1.619	0.996	1537.3	6.0819	2.10	1.96
475	1.636	0.996	1547.9	6.1043	2.12	1.98
480	1.654	0.996	1558.5	6.1265	2.14	2.00
485	1.671	0.996	1569.3	6.1487	2.15	2.01
490	1.689	0.996	1580.1	6.1709	2.17	2.03
495	1.706	0.997	1590.9	6.1930	2.18	2.04
500	1.724	0.997	1601.9	6.2149	2.20	2.06
505	1.741	0.997	1612.9	6.2369	2.21	2.07
510	1.758	0.997	1624.0	6.2587	2.23	2.09
515	1.776	0.997	1635.1	6.2805	2.24	2.10
520	1.793	0.997	1646.4	6.3022	2.26	2.12
525	1.810	0.997	1657.7	6.3239	2.27	2.13
530	1.828	0.997	1669.1	6.3455	2.28	2.14
535	1.845	0.997	1680.5	6.3670	2.30	2.16
540	1.863	0.997	1692.1	6.3884	2.31	2.17
545	1.880	0.997	1703.7	6.4098	2.33	2.19
550	1.897	0.997	1715.3	6.4311	2.34	2.20
555	1.915	0.997	1727.1	6.4524	2.36	2.22
560	1.932	0.998	1738.9	6.4736	2.37	2.23
565	1.950	0.998	1750.8	6.4947	2.38	2.24
570	1.967	0.998	1762.7	6.5158	2.40	2.26
575	1.984	0.998	1774.7	6.5367	2.41	2.27
580	2.002	0.998	1786.8	6.5577	2.42	2.28

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 50 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
338.78*	0.001345†	0.001†	592.03†	3.3783†	3.15†	3.03†
338.78†	0.9153*	0.976*	1287.0*	5.4302*	1.72*	1.54*
270	0.001246	0.002	407.03	2.7717	2.29	1.54
275	0.001252	0.002	418.59	2.8141	2.33	1.67
280	0.001257	0.002	430.34	2.8565	2.37	1.81
285	0.001263	0.002	442.35	2.8990	2.43	1.94
290	0.001270	0.002	454.63	2.9417	2.49	2.07
295	0.001276	0.002	467.22	2.9847	2.55	2.20
300	0.001283	0.002	480.14	3.0282	2.62	2.32
305	0.001290	0.002	493.40	3.0720	2.69	2.44
310	0.001297	0.002	507.01	3.1162	2.75	2.55
315	0.001305	0.001	520.96	3.1609	2.83	2.65
320	0.001313	0.001	535.26	3.2059	2.90	2.74
325	0.001321	0.001	549.91	3.2514	2.96	2.83
330	0.001329	0.001	564.90	3.2971	3.03	2.91
335	0.001338	0.001	580.23	3.3432	3.10	2.98
340	0.9192	0.977	1289.1	5.4364	1.72	1.54
345	0.9347	0.979	1297.8	5.4617	1.73	1.56
350	0.9501	0.981	1306.5	5.4867	1.75	1.58
355	0.9652	0.983	1315.2	5.5115	1.76	1.59
360	0.9803	0.984	1324.1	5.5362	1.77	1.61
365	0.9951	0.985	1332.9	5.5608	1.79	1.63
370	1.010	0.986	1341.9	5.5852	1.80	1.64
375	1.025	0.987	1351.0	5.6094	1.82	1.66
380	1.039	0.988	1360.1	5.6336	1.83	1.68
385	1.054	0.989	1369.3	5.6576	1.84	1.69
390	1.068	0.990	1378.5	5.6815	1.86	1.71
395	1.082	0.990	1387.9	5.7053	1.88	1.73
400	1.097	0.991	1397.3	5.7290	1.89	1.74
405	1.111	0.991	1406.8	5.7525	1.91	1.76
410	1.125	0.992	1416.3	5.7760	1.92	1.78
415	1.139	0.992	1426.0	5.7994	1.94	1.79
420	1.154	0.993	1435.7	5.8227	1.95	1.81
425	1.168	0.993	1445.5	5.8459	1.97	1.82
430	1.182	0.993	1455.4	5.8690	1.98	1.84
435	1.196	0.994	1465.3	5.8920	2.00	1.86
440	1.210	0.994	1475.4	5.9149	2.01	1.87
445	1.224	0.994	1485.5	5.9378	2.03	1.89
450	1.238	0.994	1495.6	5.9605	2.04	1.90
455	1.252	0.994	1505.9	5.9832	2.06	1.92
460	1.266	0.995	1516.2	6.0058	2.08	1.93
465	1.280	0.995	1526.7	6.0283	2.09	1.95
470	1.294	0.995	1537.2	6.0508	2.11	1.97
475	1.308	0.995	1547.7	6.0731	2.12	1.98
480	1.322	0.995	1558.4	6.0954	2.14	2.00
485	1.336	0.995	1569.1	6.1177	2.15	2.01
490	1.350	0.996	1579.9	6.1398	2.17	2.03
495	1.364	0.996	1590.8	6.1619	2.18	2.04
500	1.378	0.996	1601.7	6.1839	2.20	2.06
505	1.392	0.996	1612.7	6.2058	2.21	2.07
510	1.405	0.996	1623.8	6.2277	2.23	2.09
515	1.419	0.996	1635.0	6.2495	2.24	2.10
520	1.433	0.996	1646.2	6.2712	2.26	2.12
525	1.447	0.996	1657.5	6.2928	2.27	2.13
530	1.461	0.996	1668.9	6.3144	2.28	2.14
535	1.475	0.996	1680.4	6.3359	2.30	2.16
540	1.489	0.997	1691.9	6.3574	2.31	2.17
545	1.503	0.997	1703.5	6.3788	2.33	2.19
550	1.517	0.997	1715.2	6.4001	2.34	2.20
555	1.531	0.997	1726.9	6.4213	2.35	2.21
560	1.545	0.997	1738.7	6.4425	2.37	2.23
565	1.559	0.997	1750.6	6.4636	2.38	2.24
570	1.573	0.997	1762.6	6.4847	2.40	2.26
575	1.587	0.997	1774.6	6.5057	2.41	2.27
580	1.601	0.997	1786.6	6.5266	2.42	2.28

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 70 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
346.43*	0.001359†	0.002†	616.53†	3.4497†	3.25†	3.12†
346.43†	0.6631*	0.968*	1297.4*	5.4152*	1.80*	1.60*
270	0.001246	0.002	407.05	2.7717	2.29	1.54
275	0.001252	0.002	418.60	2.8141	2.33	1.67
280	0.001257	0.002	430.36	2.8565	2.37	1.81
285	0.001263	0.002	442.37	2.8989	2.43	1.94
290	0.001270	0.002	454.65	2.9417	2.49	2.07
295	0.001276	0.002	467.24	2.9847	2.55	2.20
300	0.001283	0.002	480.16	3.0281	2.62	2.32
305	0.001290	0.002	493.42	3.0720	2.69	2.44
310	0.001297	0.002	507.02	3.1162	2.76	2.55
315	0.001305	0.002	520.97	3.1609	2.83	2.65
320	0.001312	0.002	535.28	3.2059	2.90	2.74
325	0.001321	0.002	549.92	3.2513	2.96	2.83
330	0.001329	0.002	564.92	3.2971	3.03	2.91
335	0.001338	0.002	580.24	3.3432	3.10	2.98
340	0.001347	0.002	595.90	3.3896	3.16	3.04
345	0.001357	0.002	611.89	3.4363	3.23	3.10
350	0.6715	0.971	1303.8	5.4337	1.81	1.61
355	0.6831	0.974	1312.9	5.4593	1.81	1.63
360	0.6944	0.976	1321.9	5.4847	1.82	1.64
365	0.7056	0.978	1331.0	5.5098	1.83	1.65
370	0.7166	0.980	1340.2	5.5347	1.84	1.67
375	0.7275	0.982	1349.4	5.5594	1.85	1.68
380	0.7383	0.983	1358.7	5.5840	1.86	1.70
385	0.7489	0.984	1368.0	5.6083	1.87	1.71
390	0.7595	0.985	1377.4	5.6325	1.88	1.73
395	0.7700	0.986	1386.8	5.6566	1.89	1.74
400	0.7804	0.987	1396.3	5.6805	1.91	1.75
405	0.7907	0.988	1405.9	5.7043	1.92	1.77
410	0.8010	0.989	1415.5	5.7279	1.93	1.79
415	0.8113	0.989	1425.2	5.7515	1.95	1.80
420	0.8215	0.990	1435.0	5.7749	1.96	1.82
425	0.8317	0.990	1444.8	5.7982	1.98	1.83
430	0.8419	0.991	1454.8	5.8214	1.99	1.85
435	0.8520	0.991	1464.8	5.8445	2.01	1.86
440	0.8621	0.991	1474.8	5.8675	2.02	1.88
445	0.8722	0.992	1485.0	5.8904	2.04	1.89
450	0.8822	0.992	1495.2	5.9133	2.05	1.91
455	0.8923	0.992	1505.5	5.9360	2.07	1.92
460	0.9023	0.993	1515.8	5.9587	2.08	1.94
465	0.9124	0.993	1526.3	5.9812	2.09	1.95
470	0.9224	0.993	1536.8	6.0037	2.11	1.97
475	0.9324	0.993	1547.4	6.0261	2.12	1.98
480	0.9424	0.993	1558.0	6.0484	2.14	2.00
485	0.9524	0.994	1568.8	6.0707	2.15	2.01
490	0.9624	0.994	1579.6	6.0928	2.17	2.03
495	0.9724	0.994	1590.4	6.1149	2.18	2.04
500	0.9823	0.994	1601.4	6.1369	2.20	2.06
505	0.9923	0.994	1612.4	6.1589	2.21	2.07
510	1.002	0.994	1623.5	6.1807	2.23	2.09
515	1.012	0.995	1634.7	6.2025	2.24	2.10
520	1.022	0.995	1645.9	6.2243	2.26	2.11
525	1.032	0.995	1657.2	6.2459	2.27	2.13
530	1.042	0.995	1668.6	6.2675	2.28	2.14
535	1.052	0.995	1680.1	6.2890	2.30	2.16
540	1.062	0.995	1691.6	6.3104	2.31	2.17
545	1.072	0.995	1703.2	6.3318	2.33	2.19
550	1.082	0.995	1714.9	6.3531	2.34	2.20
555	1.092	0.996	1726.6	6.3744	2.35	2.21
560	1.102	0.996	1738.4	6.3955	2.37	2.23
565	1.112	0.996	1750.3	6.4166	2.38	2.24
570	1.122	0.996	1762.2	6.4377	2.39	2.25
575	1.132	0.996	1774.2	6.4587	2.41	2.27
580	1.142	0.996	1786.3	6.4796	2.42	2.28

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 100 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
355.06*	0.001377†	0.003†	645.00†	3.5307†	3.35†	3.19†
355.06†	0.4703*	0.957*	1308.5*	5.3993*	1.93*	1.70*
270	0.001246	0.003	407.08	2.7717	2.29	1.54
275	0.001252	0.003	418.64	2.8141	2.33	1.67
280	0.001257	0.003	430.39	2.8564	2.38	1.81
285	0.001263	0.003	442.39	2.8989	2.43	1.94
290	0.001269	0.003	454.68	2.9416	2.49	2.07
295	0.001276	0.003	467.27	2.9847	2.55	2.20
300	0.001283	0.003	480.19	3.0281	2.62	2.32
305	0.001290	0.003	493.45	3.0719	2.69	2.44
310	0.001297	0.003	507.05	3.1162	2.76	2.54
315	0.001304	0.003	521.00	3.1608	2.82	2.65
320	0.001312	0.003	535.30	3.2059	2.89	2.74
325	0.001320	0.003	549.95	3.2513	2.96	2.83
330	0.001329	0.003	564.94	3.2970	3.03	2.91
335	0.001338	0.003	580.27	3.3431	3.10	2.98
340	0.001347	0.003	595.93	3.3895	3.16	3.04
345	0.001356	0.003	611.91	3.4362	3.23	3.10
350	0.001366	0.003	628.20	3.4831	3.29	3.15
355	0.001377	0.003	644.81	3.5302	3.35	3.19
360	0.4791	0.962	1318.0	5.4258	1.92	1.70
365	0.4877	0.966	1327.5	5.4523	1.91	1.71
370	0.4960	0.969	1337.1	5.4782	1.91	1.71
375	0.5042	0.972	1346.6	5.5039	1.91	1.72
380	0.5122	0.974	1356.2	5.5292	1.91	1.73
385	0.5200	0.976	1365.8	5.5542	1.92	1.74
390	0.5277	0.978	1375.4	5.5790	1.92	1.75
395	0.5354	0.980	1385.0	5.6036	1.93	1.77
400	0.5429	0.981	1394.7	5.6279	1.94	1.78
405	0.5503	0.982	1404.4	5.6521	1.95	1.79
410	0.5577	0.983	1414.2	5.6760	1.96	1.80
415	0.5651	0.984	1424.0	5.6999	1.97	1.82
420	0.5724	0.985	1433.9	5.7235	1.98	1.83
425	0.5796	0.986	1443.8	5.7471	1.99	1.84
430	0.5868	0.986	1453.8	5.7705	2.01	1.86
435	0.5940	0.987	1463.9	5.7937	2.02	1.87
440	0.6012	0.988	1474.0	5.8169	2.03	1.88
445	0.6083	0.988	1484.2	5.8399	2.05	1.90
450	0.6154	0.988	1494.5	5.8628	2.06	1.91
455	0.6225	0.989	1504.8	5.8857	2.07	1.93
460	0.6296	0.989	1515.2	5.9084	2.09	1.94
465	0.6366	0.990	1525.7	5.9310	2.10	1.96
470	0.6437	0.990	1536.2	5.9536	2.11	1.97
475	0.6507	0.990	1546.8	5.9760	2.13	1.99
480	0.6578	0.991	1557.5	5.9984	2.14	2.00
485	0.6648	0.991	1568.2	6.0207	2.16	2.01
490	0.6718	0.991	1579.1	6.0428	2.17	2.03
495	0.6789	0.991	1590.0	6.0650	2.18	2.04
500	0.6859	0.992	1600.9	6.0870	2.20	2.06
505	0.6929	0.992	1611.9	6.1089	2.21	2.07
510	0.6999	0.992	1623.1	6.1308	2.23	2.09
515	0.7069	0.992	1634.2	6.1526	2.24	2.10
520	0.7139	0.992	1645.5	6.1743	2.26	2.11
525	0.7209	0.993	1656.8	6.1960	2.27	2.13
530	0.7280	0.993	1668.2	6.2176	2.28	2.14
535	0.7350	0.993	1679.6	6.2391	2.30	2.16
540	0.7420	0.993	1691.1	6.2605	2.31	2.17
545	0.7490	0.993	1702.7	6.2819	2.33	2.18
550	0.7560	0.994	1714.4	6.3032	2.34	2.20
555	0.7630	0.994	1726.1	6.3244	2.35	2.21
560	0.7701	0.994	1737.9	6.3456	2.37	2.22
565	0.7771	0.994	1749.8	6.3667	2.38	2.24
570	0.7841	0.994	1761.7	6.3877	2.39	2.25
575	0.7911	0.994	1773.7	6.4087	2.41	2.26
580	0.7982	0.995	1785.8	6.4295	2.42	2.28

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 150 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
365.57*	0.001400†	0.004†	680.91†	3.6302†	3.47†	3.26†
365.57†	0.3173*	0.941*	1321.0*	5.3808*	2.13*	1.84*
270	0.001246	0.005	407.13	2.7716	2.29	1.54
275	0.001252	0.005	418.68	2.8140	2.33	1.67
280	0.001257	0.005	430.44	2.8564	2.38	1.81
285	0.001263	0.005	442.44	2.8988	2.43	1.94
290	0.001269	0.005	454.72	2.9416	2.49	2.07
295	0.001276	0.005	467.31	2.9846	2.55	2.20
300	0.001283	0.005	480.23	3.0280	2.62	2.32
305	0.001289	0.005	493.49	3.0719	2.69	2.43
310	0.001297	0.005	507.09	3.1161	2.75	2.54
315	0.001304	0.004	521.04	3.1607	2.82	2.65
320	0.001312	0.004	535.34	3.2058	2.89	2.74
325	0.001320	0.004	549.99	3.2512	2.96	2.83
330	0.001329	0.004	564.98	3.2970	3.03	2.90
335	0.001337	0.004	580.30	3.3431	3.10	2.97
340	0.001346	0.004	595.96	3.3894	3.16	3.04
345	0.001356	0.004	611.94	3.4361	3.23	3.09
350	0.001366	0.004	628.24	3.4830	3.29	3.14
355	0.001376	0.004	644.84	3.5301	3.35	3.19
360	0.001387	0.004	661.74	3.5774	3.41	3.23
365	0.001399	0.004	678.93	3.6248	3.47	3.26
370	0.3231	0.947	1330.4	5.4063	2.10	1.83
375	0.3295	0.953	1340.8	5.4343	2.07	1.83
380	0.3355	0.957	1351.1	5.4615	2.05	1.82
385	0.3414	0.961	1361.3	5.4882	2.03	1.82
390	0.3470	0.965	1371.4	5.5144	2.02	1.82
395	0.3525	0.968	1381.5	5.5401	2.02	1.82
400	0.3579	0.970	1391.6	5.5654	2.01	1.83
405	0.3631	0.972	1401.6	5.5904	2.01	1.83
410	0.3683	0.974	1411.7	5.6151	2.01	1.84
415	0.3734	0.976	1421.8	5.6395	2.02	1.85
420	0.3785	0.977	1431.9	5.6637	2.02	1.86
425	0.3835	0.978	1442.0	5.6877	2.03	1.87
430	0.3884	0.979	1452.2	5.7114	2.04	1.88
435	0.3933	0.980	1462.4	5.7350	2.05	1.89
440	0.3982	0.981	1472.6	5.7585	2.05	1.90
445	0.4030	0.982	1482.9	5.7817	2.06	1.91
450	0.4078	0.983	1493.3	5.8049	2.08	1.93
455	0.4126	0.983	1503.7	5.8279	2.09	1.94
460	0.4174	0.984	1514.1	5.8507	2.10	1.95
465	0.4222	0.984	1524.7	5.8735	2.11	1.96
470	0.4269	0.985	1535.3	5.8961	2.12	1.98
475	0.4317	0.985	1545.9	5.9187	2.14	1.99
480	0.4364	0.986	1556.6	5.9411	2.15	2.00
485	0.4411	0.986	1567.4	5.9634	2.16	2.02
490	0.4458	0.987	1578.2	5.9857	2.17	2.03
495	0.4506	0.987	1589.1	6.0078	2.19	2.04
500	0.4553	0.987	1600.1	6.0299	2.20	2.06
505	0.4600	0.988	1611.2	6.0519	2.22	2.07
510	0.4647	0.988	1622.3	6.0737	2.23	2.08
515	0.4694	0.988	1633.4	6.0956	2.24	2.10
520	0.4741	0.989	1644.7	6.1173	2.26	2.11
525	0.4788	0.989	1656.0	6.1389	2.27	2.13
530	0.4835	0.989	1667.4	6.1605	2.28	2.14
535	0.4883	0.989	1678.8	6.1820	2.30	2.15
540	0.4930	0.990	1690.4	6.2034	2.31	2.17
545	0.4977	0.990	1701.9	6.2248	2.32	2.18
550	0.5024	0.990	1713.6	6.2461	2.34	2.19
555	0.5071	0.991	1725.3	6.2673	2.35	2.21
560	0.5118	0.991	1737.1	6.2884	2.36	2.22
565	0.5165	0.991	1749.0	6.3095	2.38	2.23
570	0.5212	0.991	1760.9	6.3305	2.39	2.25
575	0.5260	0.992	1772.9	6.3515	2.40	2.26
580	0.5307	0.992	1784.9	6.3723	2.42	2.27

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 200 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
373.54*	0.001419†	0.005†	708.98†	3.7060†	3.56†	3.30†
373.54†	0.2394*	0.926*	1329.8*	5.3673*	2.34*	1.98*
270	0.001246	0.007	407.17	2.7716	2.30	1.54
275	0.001252	0.007	418.73	2.8139	2.33	1.67
280	0.001257	0.006	430.48	2.8563	2.37	1.81
285	0.001263	0.006	442.48	2.8988	2.43	1.94
290	0.001269	0.006	454.77	2.9415	2.49	2.07
295	0.001276	0.006	467.36	2.9846	2.55	2.20
300	0.001282	0.006	480.28	3.0280	2.62	2.32
305	0.001289	0.006	493.53	3.0718	2.69	2.43
310	0.001297	0.006	507.13	3.1160	2.76	2.54
315	0.001304	0.006	521.08	3.1607	2.82	2.64
320	0.001312	0.006	535.38	3.2057	2.89	2.74
325	0.001320	0.006	550.03	3.2511	2.96	2.82
330	0.001328	0.006	565.02	3.2969	3.03	2.90
335	0.001337	0.006	580.34	3.3430	3.10	2.97
340	0.001346	0.006	596.00	3.3894	3.16	3.03
345	0.001355	0.006	611.98	3.4360	3.23	3.09
350	0.001365	0.006	628.27	3.4829	3.29	3.14
355	0.001376	0.006	644.87	3.5300	3.35	3.18
360	0.001387	0.006	661.77	3.5773	3.41	3.22
365	0.001398	0.006	678.96	3.6247	3.47	3.25
370	0.001410	0.006	696.43	3.6722	3.52	3.28
375	0.2410	0.929	1333.1	5.3764	2.32	1.97
380	0.2463	0.937	1344.6	5.4066	2.25	1.95
385	0.2514	0.944	1355.7	5.4357	2.20	1.93
390	0.2561	0.949	1366.6	5.4638	2.16	1.91
395	0.2607	0.954	1377.3	5.4911	2.13	1.90
400	0.2651	0.958	1387.9	5.5178	2.11	1.90
405	0.2693	0.961	1398.4	5.5439	2.09	1.89
410	0.2734	0.964	1408.9	5.5696	2.08	1.89
415	0.2775	0.967	1419.3	5.5948	2.08	1.89
420	0.2814	0.969	1429.6	5.6196	2.07	1.90
425	0.2853	0.970	1440.0	5.6442	2.07	1.90
430	0.2891	0.972	1450.4	5.6684	2.07	1.91
435	0.2929	0.973	1460.7	5.6924	2.08	1.91
440	0.2966	0.975	1471.1	5.7162	2.08	1.92
445	0.3003	0.976	1481.6	5.7397	2.09	1.93
450	0.3040	0.977	1492.0	5.7631	2.09	1.94
455	0.3077	0.977	1502.5	5.7863	2.10	1.95
460	0.3113	0.978	1513.1	5.8093	2.11	1.96
465	0.3149	0.979	1523.6	5.8322	2.12	1.97
470	0.3185	0.980	1534.3	5.8549	2.13	1.98
475	0.3221	0.980	1545.0	5.8776	2.14	1.99
480	0.3257	0.981	1555.7	5.9001	2.16	2.01
485	0.3293	0.981	1566.5	5.9225	2.17	2.02
490	0.3328	0.982	1577.4	5.9448	2.18	2.03
495	0.3364	0.982	1588.3	5.9669	2.19	2.04
500	0.3400	0.983	1599.3	5.9890	2.20	2.06
505	0.3435	0.983	1610.4	6.0110	2.22	2.07
510	0.3471	0.984	1621.5	6.0329	2.23	2.08
515	0.3507	0.984	1632.7	6.0548	2.24	2.10
520	0.3542	0.985	1643.9	6.0765	2.26	2.11
525	0.3578	0.985	1655.2	6.0982	2.27	2.12
530	0.3613	0.986	1666.6	6.1197	2.28	2.14
535	0.3649	0.986	1678.0	6.1412	2.30	2.15
540	0.3685	0.986	1689.6	6.1626	2.31	2.16
545	0.3720	0.987	1701.1	6.1840	2.32	2.18
550	0.3756	0.987	1712.8	6.2053	2.34	2.19
555	0.3791	0.988	1724.5	6.2265	2.35	2.20
560	0.3827	0.988	1736.3	6.2476	2.36	2.22
565	0.3863	0.988	1748.1	6.2687	2.38	2.23
570	0.3898	0.989	1760.0	6.2897	2.39	2.24
575	0.3934	0.989	1772.0	6.3106	2.40	2.26
580	0.3970	0.989	1784.1	6.3314	2.42	2.27

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 300 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
385.61*	0.001451†	0.008†	752.78†	3.8210†	3.69†	3.33†
385.61†	0.1602*	0.901*	1341.8*	5.3479*	2.75*	2.25*
270	0.001246	0.010	407.27	2.7714	2.29	1.54
275	0.001252	0.010	418.82	2.8138	2.33	1.67
280	0.001257	0.010	430.58	2.8562	2.37	1.81
285	0.001263	0.010	442.58	2.8987	2.43	1.94
290	0.001269	0.009	454.86	2.9414	2.49	2.07
295	0.001276	0.009	467.45	2.9844	2.55	2.20
300	0.001282	0.009	480.36	3.0278	2.62	2.32
305	0.001289	0.009	493.62	3.0716	2.69	2.43
310	0.001296	0.009	507.22	3.1159	2.75	2.54
315	0.001304	0.009	521.16	3.1605	2.82	2.64
320	0.001311	0.009	535.46	3.2055	2.89	2.73
325	0.001319	0.009	550.11	3.2509	2.96	2.82
330	0.001327	0.009	565.09	3.2967	3.03	2.89
335	0.001336	0.009	580.42	3.3428	3.10	2.96
340	0.001345	0.009	596.07	3.3892	3.16	3.03
345	0.001354	0.009	612.05	3.4358	3.23	3.08
350	0.001364	0.008	628.34	3.4827	3.29	3.13
355	0.001375	0.008	644.93	3.5298	3.35	3.18
360	0.001386	0.008	661.83	3.5770	3.41	3.21
365	0.001397	0.008	679.01	3.6244	3.47	3.25
370	0.001409	0.008	696.48	3.6720	3.52	3.28
375	0.001422	0.008	714.22	3.7196	3.58	3.30
380	0.001435	0.008	732.24	3.7673	3.63	3.32
385	0.001449	0.008	750.52	3.8151	3.68	3.33
390	0.1640	0.912	1353.6	5.3782	2.61	2.19
395	0.1679	0.922	1366.3	5.4107	2.49	2.14
400	0.1715	0.930	1378.6	5.4415	2.41	2.09
405	0.1749	0.937	1390.4	5.4709	2.34	2.06
410	0.1781	0.942	1402.0	5.4992	2.28	2.03
415	0.1812	0.947	1413.3	5.5267	2.24	2.01
420	0.1841	0.951	1424.4	5.5533	2.21	2.00
425	0.1869	0.954	1435.4	5.5794	2.19	1.99
430	0.1897	0.957	1446.3	5.6049	2.17	1.98
435	0.1924	0.959	1457.1	5.6299	2.16	1.97
440	0.1950	0.961	1467.9	5.6545	2.15	1.97
445	0.1976	0.963	1478.6	5.6787	2.14	1.97
450	0.2001	0.964	1489.3	5.7027	2.14	1.98
455	0.2027	0.966	1500.1	5.7263	2.14	1.98
460	0.2051	0.967	1510.8	5.7497	2.14	1.98
465	0.2076	0.968	1521.5	5.7729	2.15	1.99
470	0.2101	0.969	1532.3	5.7960	2.15	2.00
475	0.2125	0.970	1543.0	5.8188	2.16	2.01
480	0.2150	0.971	1553.9	5.8415	2.17	2.02
485	0.2174	0.972	1564.7	5.8640	2.18	2.02
490	0.2198	0.973	1575.7	5.8864	2.19	2.04
495	0.2222	0.974	1586.6	5.9086	2.20	2.05
500	0.2247	0.974	1597.6	5.9308	2.21	2.06
505	0.2271	0.975	1608.7	5.9528	2.22	2.07
510	0.2295	0.976	1619.8	5.9748	2.23	2.08
515	0.2319	0.976	1631.0	5.9966	2.24	2.09
520	0.2343	0.977	1642.3	6.0184	2.26	2.11
525	0.2367	0.978	1653.6	6.0400	2.27	2.12
530	0.2391	0.978	1665.0	6.0616	2.28	2.13
535	0.2415	0.979	1676.4	6.0831	2.30	2.14
540	0.2440	0.980	1687.9	6.1045	2.31	2.16
545	0.2464	0.980	1699.5	6.1258	2.32	2.17
550	0.2488	0.981	1711.2	6.1471	2.33	2.18
555	0.2512	0.981	1722.9	6.1683	2.35	2.20
560	0.2536	0.982	1734.6	6.1894	2.36	2.21
565	0.2560	0.983	1746.5	6.2105	2.37	2.22
570	0.2584	0.983	1758.4	6.2314	2.39	2.24
575	0.2608	0.984	1770.4	6.2523	2.40	2.25
580	0.2632	0.984	1782.4	6.2732	2.41	2.26

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 400 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
394.84*	0.001478†	0.011†	787.30†	3.9091†	3.79†	3.34†
394.84†	0.1202*	0.880*	1350.5*	5.3345*	3.14*	2.50*
270	0.001246	0.013	407.37	2.7713	2.29	1.54
275	0.001252	0.013	418.92	2.8137	2.33	1.67
280	0.001257	0.013	430.67	2.8561	2.37	1.81
285	0.001263	0.013	442.67	2.8985	2.43	1.94
290	0.001269	0.013	454.95	2.9413	2.49	2.07
295	0.001275	0.013	467.54	2.9843	2.55	2.20
300	0.001282	0.012	480.45	3.0277	2.62	2.32
305	0.001289	0.012	493.70	3.0715	2.69	2.43
310	0.001296	0.012	507.30	3.1157	2.75	2.54
315	0.001303	0.012	521.25	3.1604	2.82	2.64
320	0.001311	0.012	535.54	3.2054	2.89	2.73
325	0.001319	0.012	550.19	3.2508	2.96	2.81
330	0.001327	0.012	565.17	3.2965	3.03	2.89
335	0.001335	0.012	580.49	3.3426	3.10	2.96
340	0.001344	0.011	596.14	3.3890	3.16	3.02
345	0.001354	0.011	612.11	3.4356	3.23	3.07
350	0.001363	0.011	628.40	3.4825	3.29	3.12
355	0.001374	0.011	644.99	3.5296	3.35	3.17
360	0.001385	0.011	661.88	3.5768	3.41	3.21
365	0.001396	0.011	679.06	3.6242	3.46	3.24
370	0.001408	0.011	696.53	3.6717	3.52	3.27
375	0.001421	0.011	714.27	3.7193	3.58	3.29
380	0.001434	0.011	732.28	3.7671	3.63	3.31
385	0.001448	0.011	750.56	3.8148	3.68	3.33
390	0.001463	0.011	769.10	3.8627	3.73	3.34
395	0.1203	0.880	1351.0	5.3358	3.13	2.50
400	0.1239	0.895	1366.0	5.3736	2.89	2.39
405	0.1271	0.907	1380.0	5.4084	2.72	2.30
410	0.1300	0.917	1393.2	5.4408	2.58	2.24
415	0.1327	0.924	1405.9	5.4715	2.49	2.18
420	0.1352	0.930	1418.1	5.5008	2.41	2.14
425	0.1375	0.936	1430.0	5.5290	2.35	2.10
430	0.1398	0.940	1441.6	5.5561	2.30	2.08
435	0.1420	0.944	1453.0	5.5825	2.26	2.06
440	0.1441	0.947	1464.3	5.6082	2.24	2.04
445	0.1461	0.949	1475.4	5.6334	2.22	2.03
450	0.1481	0.952	1486.4	5.6581	2.20	2.02
455	0.1501	0.954	1497.4	5.6823	2.19	2.01
460	0.1520	0.956	1508.3	5.7062	2.18	2.01
465	0.1539	0.957	1519.3	5.7298	2.18	2.01
470	0.1558	0.959	1530.1	5.7531	2.18	2.01
475	0.1577	0.960	1541.0	5.7762	2.18	2.02
480	0.1596	0.961	1552.0	5.7990	2.19	2.02
485	0.1614	0.962	1562.9	5.8217	2.19	2.03
490	0.1633	0.964	1573.9	5.8442	2.20	2.04
495	0.1652	0.965	1584.9	5.8665	2.21	2.05
500	0.1670	0.966	1595.9	5.8888	2.21	2.06
505	0.1688	0.967	1607.0	5.9108	2.22	2.07
510	0.1707	0.968	1618.2	5.9328	2.24	2.08
515	0.1725	0.969	1629.4	5.9547	2.25	2.09
520	0.1744	0.969	1640.6	5.9764	2.26	2.10
525	0.1762	0.970	1652.0	5.9981	2.27	2.11
530	0.1780	0.971	1663.3	6.0197	2.28	2.13
535	0.1799	0.972	1674.8	6.0412	2.30	2.14
540	0.1817	0.973	1686.3	6.0626	2.31	2.15
545	0.1835	0.974	1697.9	6.0839	2.32	2.17
550	0.1854	0.974	1709.5	6.1052	2.33	2.18
555	0.1872	0.975	1721.2	6.1264	2.35	2.19
560	0.1890	0.976	1733.0	6.1475	2.36	2.21
565	0.1909	0.977	1744.8	6.1686	2.37	2.22
570	0.1927	0.978	1756.7	6.1895	2.39	2.23
575	0.1945	0.978	1768.7	6.2105	2.40	2.25
580	0.1964	0.979	1780.8	6.2313	2.41	2.26

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 500 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
402.42*	0.001502†	0.013†	816.32†	3.9815†	3.86†	3.34†
402.42†	0.09593*	0.862*	1357.5*	5.3252*	3.51*	2.73*
270	0.001246	0.017	407.47	2.7712	2.29	1.54
275	0.001252	0.016	419.02	2.8136	2.33	1.68
280	0.001257	0.016	430.77	2.8560	2.37	1.81
285	0.001263	0.016	442.76	2.8984	2.43	1.94
290	0.001269	0.016	455.04	2.9411	2.49	2.07
295	0.001275	0.016	467.63	2.9842	2.55	2.20
300	0.001282	0.015	480.54	3.0276	2.62	2.32
305	0.001289	0.015	493.79	3.0714	2.68	2.43
310	0.001296	0.015	507.39	3.1156	2.75	2.54
315	0.001303	0.015	521.33	3.1602	2.82	2.64
320	0.001310	0.015	535.62	3.2052	2.89	2.73
325	0.001318	0.015	550.27	3.2506	2.96	2.81
330	0.001326	0.015	565.25	3.2964	3.03	2.88
335	0.001335	0.014	580.57	3.3424	3.10	2.95
340	0.001343	0.014	596.21	3.3888	3.16	3.01
345	0.001353	0.014	612.18	3.4354	3.23	3.07
350	0.001363	0.014	628.47	3.4823	3.29	3.12
355	0.001373	0.014	645.05	3.5293	3.35	3.16
360	0.001384	0.014	661.94	3.5766	3.41	3.20
365	0.001395	0.014	679.12	3.6240	3.46	3.23
370	0.001407	0.014	696.58	3.6715	3.52	3.26
375	0.001420	0.014	714.31	3.7191	3.57	3.29
380	0.001433	0.014	732.32	3.7668	3.63	3.31
385	0.001447	0.014	750.59	3.8146	3.68	3.32
390	0.001462	0.014	769.13	3.8624	3.73	3.33
395	0.001477	0.014	787.93	3.9103	3.79	3.34
400	0.001494	0.013	806.99	3.9582	3.84	3.34
405	0.09762	0.871	1366.3	5.3470	3.32	2.64
410	0.1006	0.886	1382.1	5.3858	3.03	2.51
415	0.1032	0.898	1396.7	5.4213	2.83	2.41
420	0.1055	0.908	1410.5	5.4543	2.68	2.32
425	0.1077	0.916	1423.6	5.4852	2.56	2.25
430	0.1097	0.922	1436.1	5.5147	2.47	2.20
435	0.1116	0.927	1448.3	5.5428	2.40	2.16
440	0.1134	0.932	1460.2	5.5700	2.35	2.12
445	0.1152	0.935	1471.8	5.5963	2.31	2.09
450	0.1169	0.939	1483.3	5.6218	2.27	2.07
455	0.1185	0.941	1494.6	5.6468	2.25	2.06
460	0.1201	0.944	1505.8	5.6713	2.23	2.05
465	0.1217	0.946	1516.9	5.6953	2.22	2.04
470	0.1233	0.948	1527.9	5.7190	2.21	2.04
475	0.1248	0.950	1539.0	5.7423	2.20	2.03
480	0.1263	0.951	1550.0	5.7654	2.20	2.03
485	0.1279	0.953	1561.0	5.7882	2.20	2.04
490	0.1294	0.954	1572.0	5.8108	2.21	2.04
495	0.1309	0.956	1583.1	5.8333	2.21	2.05
500	0.1324	0.957	1594.2	5.8556	2.22	2.06
505	0.1339	0.958	1605.3	5.8777	2.23	2.07
510	0.1354	0.959	1616.5	5.8997	2.24	2.08
515	0.1369	0.961	1627.7	5.9216	2.25	2.09
520	0.1384	0.962	1638.9	5.9434	2.26	2.10
525	0.1399	0.963	1650.3	5.9650	2.27	2.11
530	0.1414	0.964	1661.7	5.9866	2.28	2.12
535	0.1429	0.965	1673.1	6.0081	2.30	2.14
540	0.1444	0.966	1684.6	6.0296	2.31	2.15
545	0.1459	0.967	1696.2	6.0509	2.32	2.16
550	0.1473	0.968	1707.9	6.0722	2.34	2.18
555	0.1488	0.969	1719.6	6.0934	2.35	2.19
560	0.1503	0.970	1731.4	6.1145	2.36	2.20
565	0.1518	0.971	1743.2	6.1356	2.38	2.22
570	0.1533	0.972	1755.1	6.1566	2.39	2.23
575	0.1548	0.973	1767.1	6.1775	2.40	2.24
580	0.1563	0.974	1779.2	6.1984	2.42	2.26

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 700 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
414.65*	0.001545†	0.019†	864.33†	4.0983†	3.99†	3.33†
414.65†	0.06809*	0.831*	1369.4*	5.3151*	4.14*	3.11*
270	0.001246	0.023	407.65	2.7710	2.29	1.54
275	0.001252	0.023	419.21	2.8134	2.33	1.68
280	0.001257	0.023	430.95	2.8557	2.37	1.81
285	0.001263	0.022	442.95	2.8982	2.43	1.94
290	0.001269	0.022	455.22	2.9409	2.49	2.07
295	0.001275	0.022	467.81	2.9839	2.55	2.20
300	0.001282	0.022	480.72	3.0273	2.62	2.32
305	0.001288	0.021	493.96	3.0711	2.68	2.43
310	0.001295	0.021	507.56	3.1153	2.75	2.53
315	0.001302	0.021	521.50	3.1599	2.82	2.63
320	0.001310	0.021	535.79	3.2049	2.89	2.72
325	0.001317	0.021	550.43	3.2503	2.96	2.80
330	0.001325	0.020	565.41	3.2960	3.03	2.87
335	0.001333	0.020	580.72	3.3421	3.10	2.94
340	0.001342	0.020	596.36	3.3884	3.16	3.00
345	0.001351	0.020	612.32	3.4350	3.22	3.05
350	0.001361	0.020	628.60	3.4819	3.29	3.10
355	0.001371	0.020	645.18	3.5289	3.35	3.15
360	0.001382	0.019	662.06	3.5761	3.40	3.19
365	0.001394	0.019	679.22	3.6235	3.46	3.22
370	0.001406	0.019	696.68	3.6710	3.52	3.25
375	0.001418	0.019	714.40	3.7186	3.57	3.27
380	0.001432	0.019	732.40	3.7662	3.63	3.29
385	0.001446	0.019	750.66	3.8140	3.68	3.31
390	0.001460	0.019	769.19	3.8618	3.73	3.32
395	0.001476	0.019	787.97	3.9097	3.78	3.33
400	0.001492	0.019	807.02	3.9576	3.84	3.33
405	0.001509	0.019	826.33	4.0055	3.89	3.33
410	0.001527	0.019	845.90	4.0536	3.94	3.33
415	0.06829	0.833	1370.9	5.3186	4.09	3.09
420	0.07084	0.853	1390.0	5.3643	3.58	2.86
425	0.07301	0.869	1407.0	5.4045	3.23	2.68
430	0.07491	0.881	1422.5	5.4408	2.99	2.55
435	0.07661	0.891	1436.9	5.4742	2.80	2.44
440	0.07818	0.899	1450.5	5.5054	2.66	2.35
445	0.07963	0.905	1463.6	5.5348	2.55	2.27
450	0.08100	0.911	1476.1	5.5628	2.47	2.22
455	0.08231	0.915	1488.2	5.5896	2.40	2.17
460	0.08356	0.919	1500.1	5.6156	2.35	2.13
465	0.08478	0.923	1511.8	5.6408	2.31	2.11
470	0.08597	0.926	1523.2	5.6653	2.28	2.08
475	0.08714	0.928	1534.6	5.6893	2.26	2.07
480	0.08829	0.931	1545.8	5.7129	2.24	2.06
485	0.08943	0.933	1557.0	5.7361	2.24	2.05
490	0.09056	0.935	1568.2	5.7590	2.23	2.05
495	0.09168	0.937	1579.3	5.7816	2.23	2.05
500	0.09280	0.939	1590.5	5.8040	2.23	2.06
505	0.09392	0.941	1601.7	5.8263	2.24	2.06
510	0.09503	0.943	1612.9	5.8483	2.24	2.07
515	0.09615	0.945	1624.1	5.8703	2.25	2.08
520	0.09726	0.946	1635.4	5.8921	2.26	2.09
525	0.09837	0.948	1646.7	5.9138	2.28	2.10
530	0.09948	0.950	1658.2	5.9354	2.29	2.11
535	0.1006	0.951	1669.6	5.9570	2.30	2.13
540	0.1017	0.953	1681.2	5.9785	2.32	2.14
545	0.1028	0.954	1692.8	5.9999	2.33	2.16
550	0.1039	0.956	1704.5	6.0212	2.34	2.17
555	0.1050	0.957	1716.2	6.0425	2.36	2.19
560	0.1061	0.959	1728.0	6.0637	2.37	2.20
565	0.1072	0.960	1739.9	6.0849	2.39	2.22
570	0.1083	0.961	1751.9	6.1060	2.40	2.24
575	0.1094	0.962	1764.0	6.1270	2.42	2.25
580	0.1104	0.963	1776.1	6.1480	2.43	2.27

* Value for the saturated liquid.
† Value for the saturated vapour.

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TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 1000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
428.78*	0.001604†	0.027†	921.69†	4.2332†	4.14†	3.28†
428.78†	0.04707*	0.794*	1385.6*	5.3133*	4.83*	3.52*
270	0.001246	0.033	407.95	2.7707	2.29	1.55
275	0.001251	0.033	419.50	2.8131	2.33	1.68
280	0.001257	0.032	431.23	2.8554	2.37	1.81
285	0.001263	0.032	443.22	2.8978	2.43	1.94
290	0.001269	0.032	455.50	2.9405	2.48	2.07
295	0.001275	0.031	468.08	2.9835	2.55	2.20
300	0.001281	0.031	480.98	3.0269	2.61	2.31
305	0.001288	0.031	494.23	3.0707	2.68	2.43
310	0.001294	0.030	507.82	3.1149	2.75	2.53
315	0.001301	0.030	521.76	3.1595	2.82	2.62
320	0.001308	0.030	536.04	3.2045	2.89	2.71
325	0.001316	0.029	550.67	3.2499	2.96	2.79
330	0.001323	0.029	565.64	3.2956	3.03	2.86
335	0.001332	0.029	580.95	3.3416	3.09	2.93
340	0.001340	0.028	596.58	3.3879	3.16	2.98
345	0.001349	0.028	612.53	3.4345	3.22	3.04
350	0.001359	0.028	628.80	3.4813	3.28	3.09
355	0.001369	0.028	645.36	3.5283	3.34	3.13
360	0.001380	0.028	662.23	3.5755	3.40	3.17
365	0.001391	0.028	679.39	3.6228	3.46	3.20
370	0.001403	0.027	696.82	3.6702	3.52	3.23
375	0.001416	0.027	714.54	3.7178	3.57	3.26
380	0.001429	0.027	732.52	3.7654	3.62	3.28
385	0.001443	0.027	750.77	3.8131	3.68	3.29
390	0.001458	0.027	769.27	3.8609	3.73	3.31
395	0.001473	0.027	788.04	3.9087	3.78	3.32
400	0.001489	0.027	807.07	3.9566	3.83	3.32
405	0.001506	0.027	826.35	4.0045	3.88	3.32
410	0.001524	0.027	845.89	4.0524	3.93	3.32
415	0.001544	0.027	865.70	4.1005	3.99	3.31
420	0.001564	0.027	885.78	4.1485	4.04	3.31
425	0.001586	0.027	906.12	4.1967	4.10	3.29
430	0.04761	0.800	1391.3	5.3267	4.60	3.43
435	0.04954	0.823	1412.5	5.3756	3.92	3.11
440	0.05114	0.840	1430.9	5.4176	3.47	2.87
445	0.05253	0.853	1447.4	5.4549	3.15	2.68
450	0.05377	0.864	1462.5	5.4888	2.92	2.53
455	0.05489	0.872	1476.7	5.5200	2.75	2.42
460	0.05594	0.879	1490.1	5.5493	2.61	2.32
465	0.05692	0.885	1502.9	5.5770	2.51	2.25
470	0.05787	0.890	1515.2	5.6034	2.43	2.19
475	0.05878	0.894	1527.2	5.6288	2.37	2.14
480	0.05966	0.898	1538.9	5.6534	2.33	2.11
485	0.06053	0.902	1550.5	5.6773	2.30	2.08
490	0.06138	0.905	1561.9	5.7007	2.27	2.07
495	0.06223	0.909	1573.3	5.7237	2.26	2.06
500	0.06307	0.912	1584.5	5.7464	2.25	2.05
505	0.06391	0.915	1595.8	5.7688	2.25	2.05
510	0.06474	0.918	1607.1	5.7911	2.26	2.06
515	0.06557	0.920	1618.4	5.8131	2.26	2.06
520	0.06640	0.923	1629.7	5.8350	2.27	2.08
525	0.06723	0.926	1641.1	5.8568	2.29	2.09
530	0.06805	0.928	1652.6	5.8785	2.30	2.10
535	0.06888	0.931	1664.1	5.9002	2.31	2.12
540	0.06970	0.933	1675.7	5.9218	2.33	2.14
545	0.07051	0.935	1687.4	5.9434	2.35	2.16
550	0.07133	0.937	1699.2	5.9649	2.37	2.18
555	0.07213	0.939	1711.1	5.9864	2.38	2.20
560	0.07294	0.941	1723.0	6.0079	2.40	2.22
565	0.07373	0.943	1735.1	6.0293	2.42	2.24
570	0.07453	0.945	1747.3	6.0507	2.44	2.26
575	0.07531	0.947	1759.5	6.0721	2.46	2.28
580	0.07609	0.948	1771.8	6.0934	2.47	2.30

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 1500 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
446.53*	0.001697†	0.041†	996.44†	4.4021†	4.33†	3.20†
446.53†	0.03044*	0.739*	1409.8*	5.3254*	5.48*	3.85*
270	0.001246	0.050	408.42	2.7702	2.29	1.55
275	0.001251	0.049	419.97	2.8125	2.33	1.68
280	0.001257	0.049	431.71	2.8548	2.37	1.81
285	0.001262	0.048	443.69	2.8973	2.42	1.95
290	0.001268	0.047	455.96	2.9399	2.48	2.07
295	0.001274	0.047	468.53	2.9829	2.55	2.20
300	0.001280	0.046	481.43	3.0263	2.61	2.31
305	0.001287	0.046	494.67	3.0700	2.68	2.42
310	0.001293	0.045	508.26	3.1142	2.75	2.52
315	0.001300	0.045	522.19	3.1588	2.82	2.61
320	0.001307	0.044	536.47	3.2038	2.89	2.70
325	0.001314	0.044	551.09	3.2491	2.96	2.78
330	0.001321	0.043	566.05	3.2948	3.03	2.84
335	0.001329	0.043	581.34	3.3408	3.09	2.91
340	0.001337	0.043	596.96	3.3870	3.15	2.96
345	0.001346	0.042	612.89	3.4336	3.22	3.01
350	0.001356	0.042	629.13	3.4803	3.28	3.06
355	0.001366	0.042	645.68	3.5272	3.34	3.10
360	0.001376	0.041	662.52	3.5744	3.40	3.14
365	0.001388	0.041	679.65	3.6216	3.46	3.18
370	0.001399	0.041	697.07	3.6690	3.51	3.21
375	0.001412	0.041	714.76	3.7165	3.57	3.23
380	0.001425	0.041	732.72	3.7641	3.62	3.25
385	0.001439	0.041	750.94	3.8117	3.67	3.27
390	0.001454	0.040	769.42	3.8594	3.72	3.28
395	0.001469	0.040	788.16	3.9071	3.77	3.29
400	0.001485	0.040	807.15	3.9549	3.82	3.30
405	0.001502	0.040	826.39	4.0027	3.87	3.30
410	0.001520	0.040	845.89	4.0506	3.93	3.30
415	0.001539	0.040	865.65	4.0985	3.98	3.30
420	0.001559	0.040	885.67	4.1464	4.03	3.29
425	0.001581	0.040	905.95	4.1944	4.08	3.28
430	0.001604	0.040	926.50	4.2425	4.14	3.27
435	0.001629	0.041	947.32	4.2907	4.19	3.25
440	0.001657	0.041	968.43	4.3389	4.25	3.23
445	0.001687	0.041	989.82	4.3872	4.31	3.21
450	0.03154	0.760	1427.3	5.3644	4.67	3.51
455	0.03282	0.782	1448.6	5.4117	3.93	3.13
460	0.03389	0.799	1467.0	5.4518	3.45	2.85
465	0.03483	0.812	1483.3	5.4872	3.11	2.63
470	0.03567	0.823	1498.3	5.5191	2.87	2.47
475	0.03644	0.832	1512.1	5.5485	2.69	2.34
480	0.03717	0.840	1525.2	5.5759	2.56	2.24
485	0.03787	0.846	1537.7	5.6018	2.46	2.16
490	0.03854	0.853	1549.9	5.6267	2.39	2.11
495	0.03920	0.859	1561.7	5.6507	2.34	2.07
500	0.03984	0.864	1573.3	5.6740	2.31	2.05
505	0.04048	0.869	1584.8	5.6969	2.29	2.03
510	0.04111	0.874	1596.3	5.7195	2.29	2.03
515	0.04174	0.879	1607.7	5.7418	2.29	2.04
520	0.04237	0.883	1619.2	5.7640	2.30	2.05
525	0.04299	0.888	1630.7	5.7861	2.32	2.07
530	0.04361	0.892	1642.4	5.8082	2.34	2.09
535	0.04422	0.896	1654.1	5.8302	2.36	2.12
540	0.04483	0.900	1665.9	5.8522	2.38	2.14
545	0.04543	0.904	1677.9	5.8743	2.41	2.17
550	0.04602	0.907	1690.0	5.8964	2.43	2.21
555	0.04661	0.910	1702.3	5.9186	2.46	2.24
560	0.04718	0.914	1714.6	5.9408	2.49	2.27
565	0.04775	0.916	1727.1	5.9630	2.52	2.31
570	0.04831	0.919	1739.8	5.9853	2.54	2.34
575	0.04885	0.921	1752.6	6.0076	2.57	2.38
580	0.04939	0.923	1765.5	6.0300	2.59	2.41

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 2000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
460.35*	0.001791†	0.056†	1056.4†	4.5325†	4.48†	3.11†
460.35†	0.02182*	0.685*	1428.7*	5.3388*	5.88*	3.90*
270	0.001246	0.067	408.91	2.7697	2.29	1.55
275	0.001251	0.066	420.44	2.8120	2.33	1.68
280	0.001256	0.065	432.18	2.8543	2.37	1.82
285	0.001262	0.064	444.16	2.8967	2.42	1.95
290	0.001268	0.063	456.42	2.9393	2.48	2.07
295	0.001273	0.062	468.99	2.9823	2.55	2.20
300	0.001279	0.062	481.89	3.0257	2.61	2.31
305	0.001286	0.061	495.12	3.0694	2.68	2.42
310	0.001292	0.060	508.70	3.1136	2.75	2.52
315	0.001298	0.060	522.63	3.1581	2.82	2.61
320	0.001305	0.059	536.90	3.2031	2.89	2.69
325	0.001312	0.058	551.51	3.2484	2.96	2.76
330	0.001319	0.058	566.46	3.2940	3.02	2.83
335	0.001327	0.057	581.74	3.3400	3.09	2.89
340	0.001335	0.057	597.34	3.3862	3.15	2.94
345	0.001343	0.056	613.25	3.4327	3.21	2.99
350	0.001353	0.056	629.47	3.4793	3.27	3.04
355	0.001363	0.055	646.00	3.5262	3.33	3.08
360	0.001373	0.055	662.82	3.5733	3.39	3.12
365	0.001384	0.055	679.92	3.6205	3.45	3.15
370	0.001396	0.055	697.32	3.6678	3.51	3.18
375	0.001408	0.054	714.98	3.7152	3.56	3.21
380	0.001422	0.054	732.92	3.7627	3.61	3.23
385	0.001435	0.054	751.11	3.8103	3.67	3.25
390	0.001450	0.054	769.57	3.8579	3.72	3.26
395	0.001465	0.054	788.27	3.9056	3.77	3.27
400	0.001481	0.054	807.23	3.9533	3.82	3.28
405	0.001498	0.053	826.44	4.0010	3.87	3.28
410	0.001516	0.053	845.90	4.0488	3.92	3.28
415	0.001534	0.053	865.61	4.0965	3.97	3.28
420	0.001554	0.053	885.58	4.1444	4.02	3.27
425	0.001576	0.054	905.80	4.1922	4.07	3.26
430	0.001599	0.054	926.27	4.2401	4.12	3.25
435	0.001623	0.054	947.01	4.2881	4.17	3.23
440	0.001650	0.054	968.01	4.3361	4.23	3.22
445	0.001679	0.055	989.28	4.3841	4.28	3.19
450	0.001711	0.055	1010.8	4.4323	4.34	3.17
455	0.001747	0.056	1032.7	4.4806	4.40	3.14
460	0.001788	0.056	1054.9	4.5291	4.48	3.11
465	0.02302	0.716	1452.8	5.3908	4.60	3.37
470	0.02402	0.739	1473.7	5.4355	3.83	2.97
475	0.02487	0.757	1491.5	5.4732	3.33	2.68
480	0.02561	0.771	1507.2	5.5063	3.00	2.46
485	0.02629	0.784	1521.6	5.5361	2.77	2.30
490	0.02692	0.794	1535.0	5.5635	2.60	2.18
495	0.02752	0.804	1547.7	5.5894	2.49	2.10
500	0.02810	0.812	1560.0	5.6140	2.42	2.05
505	0.02866	0.821	1571.9	5.6378	2.37	2.02
510	0.02922	0.828	1583.7	5.6610	2.35	2.01
515	0.02977	0.836	1595.5	5.6839	2.35	2.01
520	0.03031	0.842	1607.2	5.7066	2.35	2.03
525	0.03084	0.849	1619.0	5.7293	2.37	2.05
530	0.03136	0.855	1631.0	5.7519	2.40	2.08
535	0.03188	0.861	1643.0	5.7745	2.43	2.12
540	0.03239	0.867	1655.3	5.7973	2.46	2.17
545	0.03289	0.872	1667.7	5.8202	2.50	2.22
550	0.03338	0.877	1680.3	5.8432	2.54	2.27
555	0.03386	0.882	1693.1	5.8664	2.58	2.32
560	0.03433	0.886	1706.1	5.8897	2.62	2.37
565	0.03478	0.890	1719.3	5.9131	2.66	2.43
570	0.03522	0.893	1732.7	5.9367	2.70	2.48
575	0.03564	0.896	1746.3	5.9605	2.74	2.54
580	0.03605	0.898	1760.1	5.9844	2.78	2.59

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 3000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
481.65*	0.002002†	0.090†	1151.8†	4.7310†	4.92†	2.98†
481.65†	0.01266*	0.570*	1444.1*	5.3362*	7.54*	3.58*
270	0.001245	0.100	409.89	2.7687	2.29	1.56
275	0.001250	0.099	421.41	2.8109	2.32	1.69
280	0.001256	0.097	433.13	2.8532	2.37	1.82
285	0.001261	0.096	445.10	2.8956	2.42	1.95
290	0.001267	0.095	457.36	2.9382	2.48	2.08
295	0.001272	0.094	469.92	2.9811	2.54	2.20
300	0.001278	0.092	482.81	3.0245	2.61	2.31
305	0.001284	0.091	496.04	3.0682	2.68	2.41
310	0.001290	0.090	509.61	3.1123	2.75	2.51
315	0.001296	0.089	523.52	3.1569	2.82	2.59
320	0.001302	0.088	537.78	3.2018	2.89	2.67
325	0.001309	0.087	552.38	3.2470	2.95	2.74
330	0.001316	0.086	567.30	3.2926	3.02	2.81
335	0.001323	0.086	582.55	3.3385	3.08	2.86
340	0.001331	0.085	598.12	3.3846	3.14	2.91
345	0.001339	0.084	613.99	3.4309	3.21	2.96
350	0.001348	0.084	630.17	3.4775	3.27	3.00
355	0.001357	0.083	646.65	3.5242	3.33	3.04
360	0.001367	0.082	663.42	3.5711	3.38	3.08
365	0.001378	0.082	680.48	3.6182	3.44	3.11
370	0.001390	0.081	697.82	3.6654	3.50	3.14
375	0.001402	0.081	715.44	3.7127	3.55	3.17
380	0.001415	0.081	733.32	3.7601	3.60	3.19
385	0.001428	0.080	751.47	3.8075	3.65	3.21
390	0.001443	0.080	769.86	3.8550	3.71	3.22
395	0.001458	0.080	788.51	3.9025	3.75	3.24
400	0.001473	0.080	807.41	3.9500	3.80	3.24
405	0.001490	0.080	826.55	3.9976	3.85	3.25
410	0.001507	0.080	845.93	4.0451	3.90	3.25
415	0.001526	0.080	865.55	4.0927	3.95	3.25
420	0.001545	0.080	885.42	4.1403	4.00	3.24
425	0.001566	0.080	905.52	4.1879	4.04	3.23
430	0.001588	0.080	925.86	4.2355	4.09	3.22
435	0.001611	0.080	946.45	4.2830	4.14	3.21
440	0.001637	0.081	967.26	4.3306	4.19	3.19
445	0.001665	0.081	988.31	4.3782	4.23	3.17
450	0.001695	0.082	1009.6	4.4258	4.28	3.15
455	0.001728	0.082	1031.1	4.4733	4.33	3.12
460	0.001764	0.083	1052.9	4.5209	4.38	3.09
465	0.001806	0.084	1074.9	4.5686	4.44	3.06
470	0.001853	0.085	1097.3	4.6165	4.52	3.03
475	0.001908	0.087	1120.2	4.6649	4.63	3.00
480	0.001976	0.089	1143.8	4.7143	4.83	2.98
485	0.01353	0.605	1465.2	5.3800	5.46	3.10
490	0.01449	0.641	1488.7	5.4282	4.12	2.64
495	0.01526	0.668	1507.4	5.4662	3.45	2.34
500	0.01592	0.690	1523.6	5.4987	3.06	2.16
505	0.01651	0.709	1538.3	5.5279	2.82	2.04
510	0.01706	0.725	1552.0	5.5549	2.69	1.99
515	0.01758	0.740	1565.2	5.5808	2.62	1.97
520	0.01808	0.754	1578.3	5.6060	2.60	1.99
525	0.01857	0.767	1591.3	5.6308	2.60	2.04
530	0.01903	0.779	1604.4	5.6556	2.63	2.10
535	0.01948	0.790	1617.6	5.6806	2.68	2.18
540	0.01991	0.800	1631.2	5.7058	2.74	2.27
545	0.02033	0.809	1645.0	5.7313	2.80	2.37
550	0.02073	0.817	1659.2	5.7571	2.87	2.47
555	0.02111	0.825	1673.7	5.7834	2.94	2.58
560	0.02148	0.832	1688.5	5.8101	3.01	2.69
565	0.02182	0.837	1703.8	5.8371	3.08	2.80
570	0.02214	0.842	1719.4	5.8646	3.16	2.91
575	0.02243	0.846	1735.3	5.8925	3.23	3.02
580	0.02270	0.849	1751.7	5.9208	3.31	3.14

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 4000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
498.00*	0.002296†	0.133†	1230.9†	4.8881†	6.66†	2.96†
498.00†	0.007652*	0.444*	1426.5*	5.2802*	17.00*	3.27*
270	0.001245	0.133	410.86	2.7677	2.28	1.57
275	0.001250	0.131	422.38	2.8099	2.32	1.70
280	0.001255	0.130	434.09	2.8521	2.37	1.83
285	0.001260	0.128	446.06	2.8945	2.42	1.96
290	0.001266	0.126	458.31	2.9371	2.48	2.08
295	0.001271	0.125	470.86	2.9800	2.54	2.20
300	0.001277	0.123	483.75	3.0233	2.61	2.31
305	0.001282	0.122	496.97	3.0670	2.68	2.41
310	0.001288	0.120	510.53	3.1111	2.75	2.50
315	0.001294	0.119	524.44	3.1556	2.82	2.58
320	0.001300	0.117	538.68	3.2005	2.88	2.66
325	0.001306	0.116	553.26	3.2457	2.95	2.73
330	0.001313	0.115	568.16	3.2912	3.01	2.79
335	0.001320	0.114	583.38	3.3370	3.08	2.84
340	0.001327	0.113	598.91	3.3830	3.14	2.89
345	0.001335	0.112	614.75	3.4293	3.20	2.94
350	0.001344	0.111	630.89	3.4757	3.26	2.98
355	0.001353	0.110	647.32	3.5223	3.32	3.01
360	0.001363	0.109	664.04	3.5691	3.37	3.05
365	0.001373	0.109	681.05	3.6160	3.43	3.08
370	0.001384	0.108	698.34	3.6631	3.49	3.11
375	0.001396	0.108	715.91	3.7102	3.54	3.13
380	0.001409	0.107	733.74	3.7574	3.59	3.15
385	0.001422	0.107	751.83	3.8047	3.64	3.17
390	0.001436	0.106	770.17	3.8521	3.69	3.19
395	0.001451	0.106	788.77	3.8994	3.74	3.20
400	0.001466	0.106	807.60	3.9468	3.79	3.21
405	0.001482	0.106	826.67	3.9942	3.84	3.22
410	0.001499	0.106	845.98	4.0416	3.88	3.22
415	0.001517	0.106	865.52	4.0890	3.93	3.22
420	0.001536	0.106	885.29	4.1363	3.98	3.21
425	0.001556	0.106	905.29	4.1837	4.02	3.21
430	0.001578	0.106	925.52	4.2310	4.07	3.20
435	0.001601	0.106	945.96	4.2782	4.11	3.18
440	0.001625	0.107	966.62	4.3255	4.15	3.17
445	0.001651	0.107	987.48	4.3726	4.19	3.15
450	0.001680	0.108	1008.5	4.4197	4.23	3.12
455	0.001711	0.109	1029.8	4.4666	4.27	3.10
460	0.001744	0.110	1051.2	4.5135	4.31	3.07
465	0.001782	0.111	1072.9	4.5603	4.35	3.04
470	0.001823	0.112	1094.7	4.6070	4.39	3.01
475	0.001871	0.114	1116.8	4.6538	4.45	2.98
480	0.001926	0.116	1139.3	4.7009	4.55	2.95
485	0.001991	0.119	1162.4	4.7487	4.70	2.93
490	0.002074	0.122	1186.5	4.7982	4.98	2.92
495	0.002190	0.128	1212.8	4.8515	5.64	2.93
500	0.008442	0.488	1448.9	5.3251	8.90	2.87
505	0.009608	0.550	1481.6	5.3901	5.18	2.37
510	0.01042	0.591	1504.2	5.4347	4.04	2.14
515	0.01108	0.622	1522.9	5.4713	3.52	2.04
520	0.01166	0.648	1539.8	5.5039	3.26	2.03
525	0.01218	0.671	1555.8	5.5345	3.15	2.08
530	0.01267	0.691	1571.4	5.5641	3.11	2.17
535	0.01312	0.709	1587.0	5.5934	3.13	2.29
540	0.01355	0.725	1602.8	5.6227	3.18	2.43
545	0.01395	0.740	1618.9	5.6524	3.26	2.59
550	0.01432	0.753	1635.4	5.6826	3.35	2.76
555	0.01467	0.764	1652.4	5.7133	3.45	2.93
560	0.01499	0.774	1669.9	5.7447	3.55	3.12
565	0.01529	0.782	1687.9	5.7767	3.66	3.30
570	0.01555	0.789	1706.5	5.8095	3.78	3.49
575	0.01579	0.794	1725.7	5.8431	3.91	3.68
580	0.01599	0.797	1745.6	5.8775	4.04	3.88

* Value for the saturated liquid.
† Value for the saturated vapour.

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 5000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
350	0.001340	0.138	631.63	3.4740	3.25	2.95
355	0.001349	0.137	648.01	3.5204	3.31	2.99
360	0.001358	0.136	664.69	3.5671	3.36	3.02
365	0.001368	0.135	681.65	3.6139	3.42	3.05
370	0.001379	0.135	698.88	3.6608	3.48	3.08
375	0.001391	0.134	716.40	3.7078	3.53	3.10
380	0.001403	0.133	734.17	3.7549	3.58	3.12
385	0.001416	0.133	752.21	3.8020	3.63	3.14
390	0.001430	0.132	770.50	3.8492	3.68	3.16
395	0.001444	0.132	789.03	3.8965	3.73	3.17
400	0.001459	0.132	807.81	3.9437	3.78	3.18
405	0.001475	0.132	826.82	3.9909	3.82	3.19
410	0.001492	0.132	846.06	4.0381	3.87	3.19
415	0.001510	0.131	865.52	4.0853	3.92	3.19
420	0.001528	0.132	885.21	4.1325	3.96	3.19
425	0.001548	0.132	905.11	4.1796	4.00	3.18
430	0.001569	0.132	925.22	4.2266	4.04	3.17
435	0.001591	0.132	945.54	4.2736	4.08	3.16
440	0.001614	0.133	966.06	4.3205	4.12	3.14
445	0.001639	0.133	986.76	4.3673	4.16	3.13
450	0.001666	0.134	1007.6	4.4139	4.19	3.10
455	0.001695	0.135	1028.7	4.4604	4.22	3.08
460	0.001727	0.136	1049.8	4.5067	4.25	3.05
465	0.001761	0.137	1071.2	4.5528	4.28	3.02
470	0.001799	0.138	1092.6	4.5987	4.30	2.99
475	0.001841	0.140	1114.2	4.6444	4.34	2.96
480	0.001888	0.142	1136.0	4.6900	4.38	2.93
485	0.001942	0.145	1158.1	4.7358	4.46	2.91
490	0.002005	0.148	1180.6	4.7821	4.58	2.89
495	0.002083	0.152	1204.0	4.8295	4.80	2.87
500	0.002185	0.158	1229.0	4.8797	5.24	2.88
505	0.002339	0.167	1257.4	4.9362	6.33	2.90
510	0.002699	0.191	1298.6	5.0174	14.33	3.00
515	0.006004	0.421	1442.6	5.2986	10.30	2.57
520	0.007109	0.494	1480.0	5.3710	5.92	2.29
525	0.007870	0.542	1506.0	5.4207	4.69	2.24
530	0.008487	0.579	1528.0	5.4625	4.19	2.31
535	0.009020	0.609	1548.4	5.5008	3.98	2.45
540	0.009495	0.635	1568.1	5.5374	3.92	2.64
545	0.009924	0.658	1587.8	5.5736	3.94	2.86
550	0.01031	0.678	1607.6	5.6099	4.01	3.10
555	0.01067	0.695	1627.9	5.6466	4.11	3.35
560	0.01099	0.709	1648.8	5.6841	4.24	3.61
565	0.01127	0.721	1670.3	5.7224	4.38	3.88
570	0.01153	0.731	1692.6	5.7616	4.54	4.16
575	0.01174	0.738	1715.7	5.8020	4.71	4.44
580	0.01192	0.743	1739.8	5.8437	4.91	4.73

Isobar at 7000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
450	0.001642	0.185	1006.1	4.4033	4.12	3.07
455	0.001669	0.186	1026.8	4.4489	4.15	3.05
460	0.001697	0.187	1047.6	4.4943	4.16	3.02
465	0.001727	0.188	1068.4	4.5394	4.17	3.00
470	0.001759	0.189	1089.3	4.5841	4.18	2.97
475	0.001794	0.191	1110.2	4.6284	4.19	2.94
480	0.001832	0.193	1131.2	4.6723	4.20	2.91
485	0.001874	0.195	1152.2	4.7158	4.21	2.88
490	0.001920	0.198	1173.3	4.7591	4.24	2.86
495	0.001972	0.202	1194.6	4.8024	4.29	2.84
500	0.002031	0.205	1216.3	4.8459	4.38	2.82
505	0.002100	0.210	1238.5	4.8901	4.53	2.81
510	0.002185	0.217	1261.7	4.9359	4.78	2.82
515	0.002294	0.225	1286.6	4.9844	5.21	2.84
520	0.002447	0.238	1314.3	5.0380	5.94	2.86
525	0.002686	0.259	1346.8	5.1002	7.17	2.89
530	0.003086	0.295	1386.3	5.1750	8.44	2.87
535	0.003639	0.344	1427.8	5.2530	7.85	2.84
540	0.004199	0.393	1464.4	5.3210	6.86	2.92
545	0.004724	0.439	1497.3	5.3817	6.37	3.14
550	0.005214	0.480	1528.6	5.4388	6.17	3.46
555	0.005665	0.516	1559.2	5.4942	6.11	3.86
560	0.006071	0.549	1589.8	5.5491	6.13	4.29
565	0.006433	0.576	1620.6	5.6039	6.22	4.73
570	0.006750	0.599	1652.1	5.6593	6.36	5.19
575	0.007025	0.618	1684.3	5.7157	6.55	5.66
580	0.007258	0.633	1717.6	5.7734	6.78	6.13

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 10 000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
450	0.001612	0.259	1004.5	4.3888	4.05	3.03
455	0.001635	0.260	1024.8	4.4336	4.06	3.01
460	0.001660	0.261	1045.1	4.4781	4.07	2.99
465	0.001686	0.262	1065.5	4.5221	4.07	2.96
470	0.001714	0.264	1085.9	4.5657	4.07	2.94
475	0.001743	0.265	1106.2	4.6087	4.06	2.91
480	0.001773	0.267	1126.4	4.6511	4.05	2.89
485	0.001806	0.269	1146.7	4.6930	4.03	2.86
490	0.001841	0.272	1166.8	4.7343	4.02	2.84
495	0.001878	0.274	1186.9	4.7751	4.02	2.82
500	0.001918	0.277	1207.0	4.8155	4.03	2.80
505	0.001961	0.281	1227.2	4.8557	4.05	2.79
510	0.002009	0.285	1247.5	4.8958	4.10	2.78
515	0.002062	0.289	1268.2	4.9361	4.17	2.79
520	0.002122	0.295	1289.3	4.9770	4.29	2.80
525	0.002191	0.302	1311.2	5.0188	4.45	2.83
530	0.002272	0.310	1334.0	5.0620	4.66	2.88
535	0.002367	0.320	1357.9	5.1068	4.90	2.94
540	0.002481	0.332	1382.9	5.1535	5.13	3.03
545	0.002613	0.347	1409.1	5.2017	5.33	3.13
550	0.002764	0.363	1436.2	5.2512	5.49	3.27
555	0.002933	0.382	1464.1	5.3017	5.69	3.45
560	0.003121	0.403	1493.3	5.3541	6.01	3.70
565	0.003335	0.427	1524.6	5.4097	6.54	4.04
570	0.003581	0.454	1559.0	5.4704	7.28	4.48
575	0.003858	0.485	1597.5	5.5376	8.10	5.04
580	0.004155	0.518	1639.9	5.6109	8.81	5.70

Isobar at 15 000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
450	0.001572	0.379	1002.9	4.3676	3.96	2.97
455	0.001592	0.379	1022.8	4.4114	3.97	2.95
460	0.001613	0.380	1042.6	4.4549	3.97	2.94
465	0.001635	0.381	1062.5	4.4978	3.97	2.92
470	0.001658	0.382	1082.3	4.5402	3.96	2.90
475	0.001681	0.384	1102.1	4.5821	3.94	2.88
480	0.001706	0.385	1121.8	4.6232	3.92	2.86
485	0.001731	0.387	1141.3	4.6638	3.90	2.84
490	0.001757	0.389	1160.7	4.7036	3.87	2.82
495	0.001785	0.391	1180.0	4.7428	3.85	2.80
500	0.001813	0.393	1199.2	4.7814	3.83	2.78
505	0.001843	0.396	1218.3	4.8194	3.81	2.77
510	0.001873	0.398	1237.3	4.8569	3.80	2.76
515	0.001906	0.401	1256.4	4.8940	3.81	2.76
520	0.001940	0.404	1275.4	4.9309	3.83	2.77
525	0.001976	0.408	1294.7	4.9677	3.87	2.78
530	0.002015	0.412	1314.2	5.0047	3.93	2.81
535	0.002058	0.417	1334.1	5.0420	4.03	2.84
540	0.002104	0.422	1354.5	5.0800	4.15	2.89
545	0.002154	0.429	1375.6	5.1189	4.30	2.96
550	0.002210	0.436	1397.5	5.1589	4.47	3.05
555	0.002271	0.444	1420.4	5.2003	4.67	3.16
560	0.002338	0.453	1444.3	5.2432	4.89	3.30
565	0.002411	0.463	1469.3	5.2877	5.13	3.46
570	0.002490	0.474	1495.6	5.3341	5.40	3.67
575	0.002574	0.485	1523.4	5.3825	5.72	3.90
580	0.002664	0.498	1553.0	5.4337	6.12	4.19

TABLE 6.3 (continued)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 20 000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
450	0.001540	0.495	1002.3	4.3488	3.90	2.92
455	0.001558	0.495	1021.8	4.3920	3.91	2.91
460	0.001577	0.495	1041.3	4.4347	3.91	2.90
465	0.001596	0.496	1060.9	4.4770	3.90	2.89
470	0.001616	0.497	1080.4	4.5187	3.89	2.87
475	0.001636	0.498	1099.8	4.5598	3.88	2.86
480	0.001657	0.499	1119.2	4.6003	3.86	2.84
485	0.001679	0.500	1138.4	4.6402	3.83	2.82
490	0.001701	0.502	1157.5	4.6793	3.80	2.81
495	0.001723	0.503	1176.4	4.7178	3.77	2.79
500	0.001746	0.505	1195.2	4.7556	3.75	2.78
505	0.001770	0.507	1213.9	4.7927	3.72	2.77
510	0.001794	0.509	1232.4	4.8292	3.70	2.76
515	0.001819	0.511	1250.9	4.8652	3.69	2.75
520	0.001845	0.513	1269.3	4.9008	3.68	2.75
525	0.001871	0.515	1287.7	4.9361	3.69	2.76
530	0.001899	0.518	1306.2	4.9712	3.71	2.77
535	0.001928	0.521	1324.9	5.0062	3.76	2.79
540	0.001958	0.524	1343.8	5.0415	3.82	2.82
545	0.001991	0.528	1363.1	5.0771	3.91	2.86
550	0.002026	0.533	1383.0	5.1133	4.03	2.92
555	0.002064	0.538	1403.4	5.1504	4.18	2.98
560	0.002105	0.543	1424.8	5.1886	4.36	3.07
565	0.002150	0.550	1447.1	5.2283	4.58	3.18
570	0.002200	0.558	1470.6	5.2698	4.84	3.31
575	0.002254	0.567	1495.5	5.3133	5.13	3.47
580	0.002312	0.576	1522.0	5.3591	5.45	3.65

Isobar at 30 000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
450	0.001490	0.718	1002.7	4.3163	3.81	2.86
455	0.001505	0.717	1021.8	4.3584	3.82	2.85
460	0.001521	0.717	1040.9	4.4002	3.82	2.85
465	0.001537	0.717	1060.0	4.4415	3.82	2.84
470	0.001553	0.717	1079.1	4.4824	3.81	2.83
475	0.001570	0.717	1098.2	4.5227	3.80	2.82
480	0.001587	0.717	1117.1	4.5624	3.78	2.81
485	0.001604	0.717	1136.0	4.6015	3.76	2.80
490	0.001622	0.718	1154.8	4.6400	3.74	2.79
495	0.001640	0.718	1173.4	4.6778	3.71	2.78
500	0.001658	0.719	1191.9	4.7149	3.68	2.77
505	0.001676	0.720	1210.2	4.7514	3.65	2.76
510	0.001694	0.720	1228.4	4.7873	3.63	2.75
515	0.001712	0.721	1246.5	4.8225	3.60	2.75
520	0.001731	0.722	1264.4	4.8572	3.58	2.74
525	0.001750	0.723	1282.3	4.8914	3.57	2.74
530	0.001769	0.724	1300.1	4.9252	3.56	2.75
535	0.001788	0.725	1317.9	4.9587	3.56	2.75
540	0.001808	0.726	1335.8	4.9919	3.58	2.76
545	0.001829	0.728	1353.7	5.0250	3.61	2.78
550	0.001850	0.729	1371.9	5.0581	3.65	2.80
555	0.001872	0.732	1390.3	5.0914	3.72	2.83
560	0.001896	0.734	1409.1	5.1251	3.80	2.87
565	0.001921	0.737	1428.3	5.1594	3.91	2.92
570	0.001948	0.741	1448.3	5.1945	4.06	2.98
575	0.001977	0.745	1469.0	5.2307	4.24	3.05
580	0.002009	0.751	1490.8	5.2684	4.47	3.14

TABLE 6.3 (concluded)
Thermodynamic properties of propan-2-ol in the single phase between 1 and 50 000 kPa

Isobar at 40 000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
450	0.001453	0.933	1004.9	4.2883	3.74	2.82
455	0.001466	0.931	1023.6	4.3297	3.75	2.82
460	0.001479	0.930	1042.4	4.3707	3.76	2.81
465	0.001493	0.929	1061.2	4.4114	3.76	2.81
470	0.001508	0.927	1080.0	4.4516	3.76	2.80
475	0.001522	0.926	1098.8	4.4914	3.75	2.80
480	0.001537	0.926	1117.5	4.5306	3.74	2.79
485	0.001552	0.925	1136.2	4.5693	3.73	2.78
490	0.001567	0.924	1154.8	4.6074	3.71	2.78
495	0.001582	0.924	1173.2	4.6450	3.68	2.77
500	0.001597	0.924	1191.6	4.6819	3.66	2.76
505	0.001612	0.923	1209.8	4.7182	3.63	2.76
510	0.001628	0.923	1227.9	4.7538	3.61	2.75
515	0.001643	0.922	1245.9	4.7889	3.58	2.75
520	0.001659	0.922	1263.8	4.8234	3.56	2.74
525	0.001674	0.922	1281.5	4.8573	3.54	2.74
530	0.001690	0.922	1299.1	4.8908	3.52	2.74
535	0.001705	0.922	1316.7	4.9238	3.51	2.74
540	0.001721	0.921	1334.2	4.9564	3.50	2.75
545	0.001737	0.921	1351.8	4.9887	3.51	2.76
550	0.001753	0.922	1369.3	5.0208	3.52	2.77
555	0.001770	0.922	1387.0	5.0527	3.54	2.78
560	0.001787	0.922	1404.8	5.0847	3.58	2.80
565	0.001804	0.923	1422.8	5.1167	3.63	2.83
570	0.001823	0.925	1441.1	5.1490	3.70	2.86
575	0.001842	0.926	1459.8	5.1816	3.79	2.89
580	0.001863	0.929	1479.1	5.2150	3.91	2.94

Isobar at 50 000 kPa

T_{68} K	v m^3/kg	Z	h kJ/kg	s kJ/kg K	c_p kJ/kg K	c_v kJ/kg K
450	0.001422	1.142	1008.2	4.2638	3.67	2.79
455	0.001434	1.139	1026.6	4.3045	3.69	2.79
460	0.001446	1.136	1045.1	4.3448	3.70	2.79
465	0.001459	1.134	1063.6	4.3849	3.71	2.79
470	0.001471	1.131	1082.1	4.4246	3.71	2.78
475	0.001484	1.129	1100.7	4.4639	3.71	2.78
480	0.001497	1.127	1119.3	4.5027	3.71	2.77
485	0.001510	1.125	1137.8	4.5411	3.70	2.77
490	0.001524	1.124	1156.3	4.5790	3.69	2.77
495	0.001537	1.122	1174.7	4.6163	3.67	2.76
500	0.001551	1.121	1193.0	4.6531	3.65	2.76
505	0.001564	1.120	1211.2	4.6893	3.63	2.75
510	0.001578	1.118	1229.2	4.7250	3.61	2.75
515	0.001592	1.117	1247.2	4.7600	3.58	2.75
520	0.001605	1.116	1265.1	4.7945	3.56	2.75
525	0.001619	1.114	1282.8	4.8285	3.54	2.75
530	0.001632	1.113	1300.4	4.8619	3.52	2.75
535	0.001646	1.112	1317.9	4.8948	3.50	2.75
540	0.001659	1.110	1335.4	4.9273	3.48	2.75
545	0.001673	1.109	1352.8	4.9593	3.48	2.75
550	0.001686	1.108	1370.2	4.9911	3.47	2.76
555	0.001700	1.107	1387.5	5.0225	3.48	2.77
560	0.001714	1.106	1405.0	5.0538	3.49	2.78
565	0.001728	1.105	1422.5	5.0849	3.51	2.80
570	0.001742	1.105	1440.1	5.1159	3.54	2.81
575	0.001757	1.105	1457.9	5.1470	3.58	2.83
580	0.001773	1.105	1475.9	5.1783	3.64	2.86

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APPENDIX A CONVERSION FACTORS

To convert from	to	multiply by
Pressure in kPa (kN/m^2)	Pressure in Pa	10^3
	Pressure in bar	10^{-2}
	Pressure in atm	$9.869\,233 \times 10^{-3}$
	Pressure in kgf/cm^2	$10.197\,16 \times 10^{-3}$
	Pressure in lbf/in^2	$145.037\,7 \times 10^{-3}$
	Pressure in mmHg	7.500 617
Specific volume in m^3/kg	Specific volume in cm^3/g	10^3
	Specific volume in ft^3/lb	16.018 45
Density in kg/m^3	Density in g/cm^3	10^{-3}
	Density in lb/ft^3	62.428×10^{-3}
Specific enthalpy in kJ/kg	Specific enthalpy in cal/g*	238.846×10^{-3}
	Specific enthalpy in Btu/lb*	429.923×10^{-3}
Specific entropy in kJ/kg K	Specific entropy in cal/g K	238.846×10^{-3}
	Specific entropy in Btu/lb °F*	238.846×10^{-3}
Specific heat capacity in kJ/kg K	Specific heat capacity in cal/g K*	238.846×10^{-3}
	Specific heat capacity in Btu/lb °F*	238.846×10^{-3}

* Note: the values used for the calorie and British Thermal Unit are International Table values.

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APPENDIX B

B1. SPECIFIC HEAT CAPACITIES OF SATURATED LIQUIDS AND VAPOURS

In addition to the constant pressure and constant volume specific heat capacities, there is another specific heat capacity which relates to the saturation line conditions of either phase.

The thermodynamic definitions of the three specific heat capacities are:

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p \ , \tag{B1.1}$$

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v \ , \tag{B1.2}$$

and

$$c_{\text{sat}} = c_p - T\left(\frac{\partial v}{\partial T}\right)_p \left(\frac{\mathrm{d}p}{\mathrm{d}T}\right)_{\text{sat}} \ , \tag{B1.3}$$

where c_p and c_v represent respectively the change in specific enthalpy with temperature at constant pressure and the change in specific internal energy with temperature at constant volume (the values commonly used in engineering design calculations), and

c_{sat} is not $(\partial h/\partial T)_{\text{sat}}$ but is the specific energy required to effect a temperature change while maintaining the fluid in a saturated state, and is the specific heat capacity commonly measured.

The three specific heat capacities are inter-related as follows:

$$c_p = c_v + T\left(\frac{\partial v}{\partial T}\right)_p \left(\frac{\partial p}{\partial T}\right)_v \ , \tag{B1.4}$$

$$\left(\frac{\mathrm{d}h}{\mathrm{d}T}\right)_{\text{sat}} = c_p + \left[v - T\left(\frac{\partial v}{\partial T}\right)_p\right]\left(\frac{\mathrm{d}p}{\mathrm{d}T}\right)_{\text{sat}} \ , \tag{B1.5}$$

$$\left(\frac{\mathrm{d}h}{\mathrm{d}T}\right)_{\text{sat}} = c_{\text{sat}} + v\left(\frac{\mathrm{d}p}{\mathrm{d}T}\right)_{\text{sat}} \ , \tag{B1.6}$$

where $(\mathrm{d}p/\mathrm{d}T)_{\text{sat}}$ represents the change in vapour pressure with temperature.

In practice, the specific heat capacity at saturation conditions cannot be measured by a constant pressure process since a phase change would accompany an energy input and the specific enthalpy of vaporisation and not the specific heat capacity would be determined. For the liquid phase, the specific heat capacities c_p and c_{sat} are in close numerical agreement at low reduced temperatures and differences can usually be neglected. For the saturated vapour, the difference between c_p and c_{sat} should not be ignored.

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89020

Thermophysical properties of propan-2-ol (isopropyl alcohol): Part 2. Thermodynamic properties at elevated pressures. ESDU 89020

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ESDU 89020 provides data for the thermodynamic properties of propan-2-ol (isopropyl alcohol) along the saturation line and in the single phase liquid, vapour and supercritical regions. The temperature range covered is from 270 K to 580 K for pressures up to 50 MPa. This Item complements and extends the information for the saturation line, ideal gas and the dilute gas available in ESDU 89017.

The tables were generated from a single Helmholtz energy function or Equation of State, from which all thermodynamic properties, at saturation as well as in the single phase, were obtained by appropriate mathematical manipulation. In this way thermodynamically consistent sets of tables were obtained. The function itself was determined empirically by data fitting to experimental *pvT* and enthalpy data.

Specific volumes, compression factors, enthalpies, entropies, isobaric and isochoric heat capacities are tabulated at saturation, and along isobars at various temperatures in the single phase. Accuracies associated with interpolation are discussed and conversion factors to facilitate use of the tables in British units are included. A computer-generated temperature-entropy chart is available from ESDU on request.

A full bibliography is given of the data sources consulted.

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